

R.V.COLLEGE OF ENGINEERING, Bengaluru

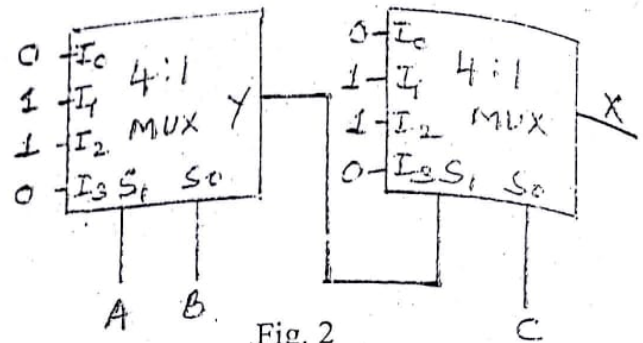
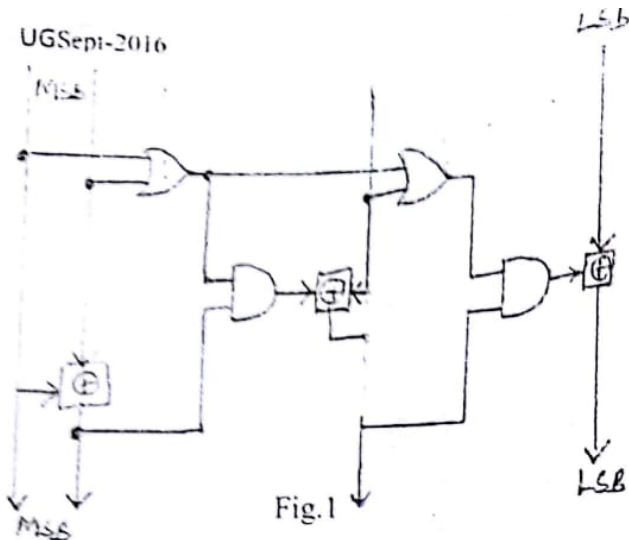
(Autonomous Institution affiliated to VTU, Belagavi)

Department of Electronics and Communication Engineering**Subject title: Analysis and Design of Digital Circuits****Subject code: (12EC34)****Semester: III****Test-1****Time : 2 hours****Maximum Marks: 65****Instructions to candidates:**

1. Answer all the questions from Part A. Part A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

PART-A

Sl.No	QUESTION	M	BT	CO
1	Realize the function $F = (A+B')(CD+E)$ using NAND gates only.	2	L1	CO1
2	Write the dual of the function $F = AB + A'C + BC = AB + A'C$.	1	L1	CO1
3	The circuit shown in the Fig. 1 is a _____ to _____ code converter.	1	L2	CO2
4	Identify essential prime implicants in the function given below. $F = \sum m(0, 2, 4, 7, 8, 10, 11, 12, 14, 15)$	1	L3	CO1
5	In Fig. 2, logic expression(X) is _____.	2	L3	CO2
6	The minimum number of 2:1 multiplexers required for realizing an 8:1 multiplexer is _____.	1	L1	CO2
7	A 74147 priority encoder has ten active-LOW inputs and four active-LOW outputs. What would be the state of the four outputs, if inputs 4 and 5 are LOW and all other inputs are HIGH?	1	L3	CO3
8	In a binary multiplier, for J multiplier bits and K multiplicand bits, hardware requirement is _____ number of AND gates and _____ numbers of K bit adders.	1	L1	CO1
9	The decoders constructed using NAND gates are _____ term generators.	1	L1	CO1
10	If the input to a T-flipflop is a 100 Hz signal, the frequency of the final output of the three T-flipflops in cascade is _____.	2	L3	CO4
11	Realize 3X8 decoder using 2X4 decoders.	2	L2	CO2



PART-B

Sl.No	QUESTION	M	BT	CO
1 a)	Simplify the following Boolean expression using K-map method. $F = \sum m(0,2,3,5,7,15,18,21,22,26,27,28,29,31) + d(4,11,16,17)$	5	L3	CO1
b)	Design a code converter that converts a decimal digit from (8421) code to BCD using suitable ICs.	5	L5	CO2
2	Design a 4-bit carry look ahead adder and list out its advantages over a ripple carry adder.	10	L3	CO2
3	Implement the following two output functions (f_1 and f_2) using suitable decoder and external gates. $f_1 = (y' + x)z$ and $f_2 = (y'z' + x'y + yz')$	10	L5	CO2
4	Obtain the Boolean function $f(a,b,c,d) = \sum m(1,5,7,8,9,13,14)$ using suitable multiplexer with a,b and c as control inputs.	10	L4	CO1
5 a)	An automobile alarm circuit is used to detect certain undesirable conditions. There are three switches used to indicate the status of the door adjacent to driver's seat, the ignition and the head lights respectively. Design the logic circuit with these three switches as the inputs, so that the alarm will be activated whenever either of the following conditions exist: (a) The head lights are on while the ignition is off. (b) The door is open while the ignition is on. $H\bar{I} + DI$	5	L3	CO4
b)	Differentiate between sequential and combinational logic circuits. Derive the characteristic equation of SR-FF and JK-FF.	5	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	6	6	1	2	6	3	6	0	0	0
	Test	Max Marks	20	25	0	5	0	5	20	10	15	0

R.V. College of Engg. B' like
Dept. of ECE

prepared by
Date 20/11/2020

Subject: Analysis & Design of Digital Circuits
Sem: III

Sub code: 12EC34

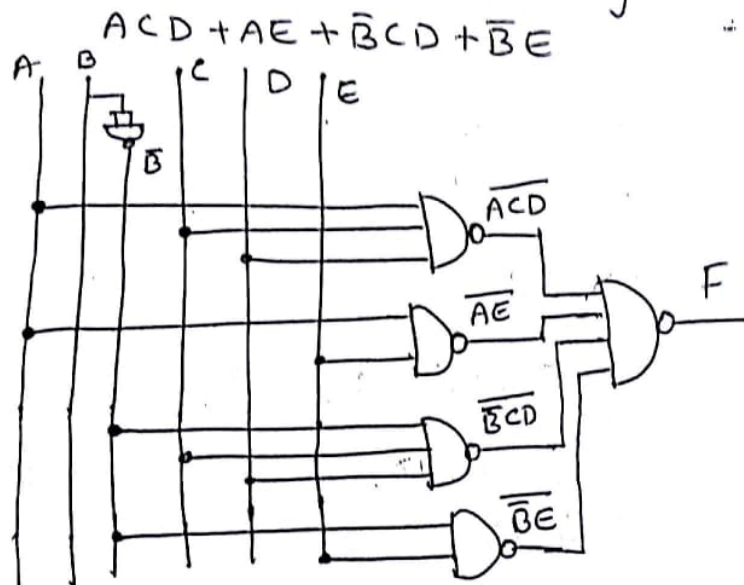
Max Marks: 65

Scheme

PART-A

Quiz

1) $F = (A + \bar{B})(CD + E)$ using NAND gates only - 2M



2) Dual of the function $F = AB + \bar{A}\bar{C} + B\bar{C} = \bar{A}B + \bar{A}\bar{C}$
 $F = (\bar{A} + \bar{B})(A + \bar{C})(\bar{B} + \bar{C}) = (\bar{A} + \bar{B})(A + \bar{C})$ - 1M.

3) BCD to Binary code. - 1M

No essential PI - 1M

$A \oplus B \oplus C$ - 2M.
 $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$

7 - 1M.

If 4 & 5 are low the o/p will be 5 i.e 1010 - 1M

JK & J-1 - 1M

9) Max.

— 1M.

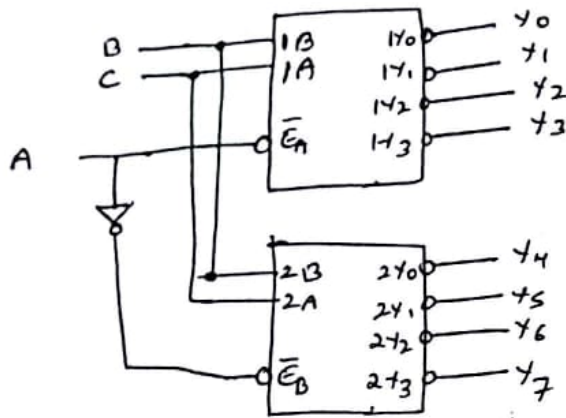
10) $100\text{Hz}/2^3 = 12.5\text{Hz}$.

— 2M.

$T = \text{Frequency}/2^N$

— 2M.

11) 3x8 decoder using 2x4 Decoders



PART-B.

1a) Simplify using 5-variable K-map.

$F = \sum m(0, 2, 3, 5, 7, 15, 18, 21, 22, 26, 27, 28, 29, 31)$
 $+ d(4, 11, 16, 17)$

ABC	000	001	011	010	100	101	111	110
DE	0	4	12	8	16	20	28	24
00	1	X			X		1	
01		5	13	9	17	21	29	25
		1			X	1	1	
11	3	7	15	11	19	23	31	27
	1	1	1	X			1	1
10	2	6	14	10	18	22	30	26
	1				1	1		1

← 4-variable K-map → ← 4-variable K-map →

000 & 100 are adjacent columns
 001 & 101 "
 011 & 111 "
 010 & 110 "

Prepared by
 Anushree

0, 2, 16, 18 - $\bar{B} \bar{C} \bar{E}$

3, 7, 11, 15 - $\bar{A} D E$

18, 22 - $A \bar{B} D \bar{E}$

28, 29 - $A B C \bar{D}$

11, 15, 27, 31 - $B D E$

5, 21 - $\bar{B} C \bar{D} E$

18, 26 - $A \bar{C} D \bar{E}$

Shashank
Date / / 20

(2)

b) Design a code converter that converts a decimal digit from (8421) code to BCD using suitable ICs.

Binary 8 4 2 1	BCD $S_4 S_3 S_2 S_1 S_0$
0 0 0 0	0 0 0 0 0
0 0 0 1	0 0 0 0 1
0 0 1 0	0 0 0 1 0
0 0 1 1	0 0 0 1 1
0 1 0 0	0 0 1 0 0
0 1 0 1	0 0 1 0 1
0 1 1 0	0 0 1 1 0
0 1 1 1	0 0 1 1 1
1 0 0 0	0 1 0 0 0
1 0 0 1	0 1 0 0 1
1 0 1 0	1 0 0 0 0
1 0 1 1	1 0 0 0 1
1 1 0 0	1 0 0 1 0
1 1 0 1	1 0 0 1 1
1 1 1 0	1 0 1 0 0
1 1 1 1	1 0 1 0 1

Theory - 1M

TT - 1M

Logic
Diagram - 3M

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01					
11		1	1	1	1
10				1	1

$$S_4 = B_3 B_2 + B_2 B_1$$

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01					
11		1	1	1	1
10				1	1

$$S_3 = B_3 \bar{B}_2 \bar{B}_1$$

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01					
11		1	1	1	1
10				1	1

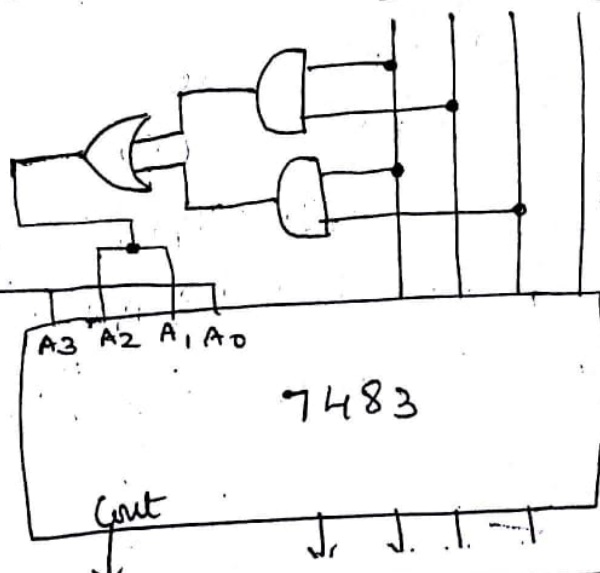
$$S_0 = B_0$$

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01					
11		1	1	1	1
10				1	1

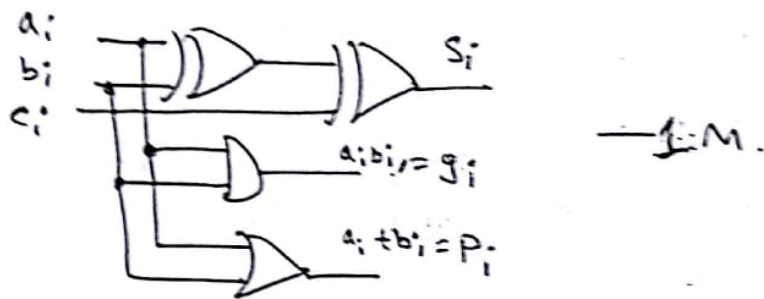
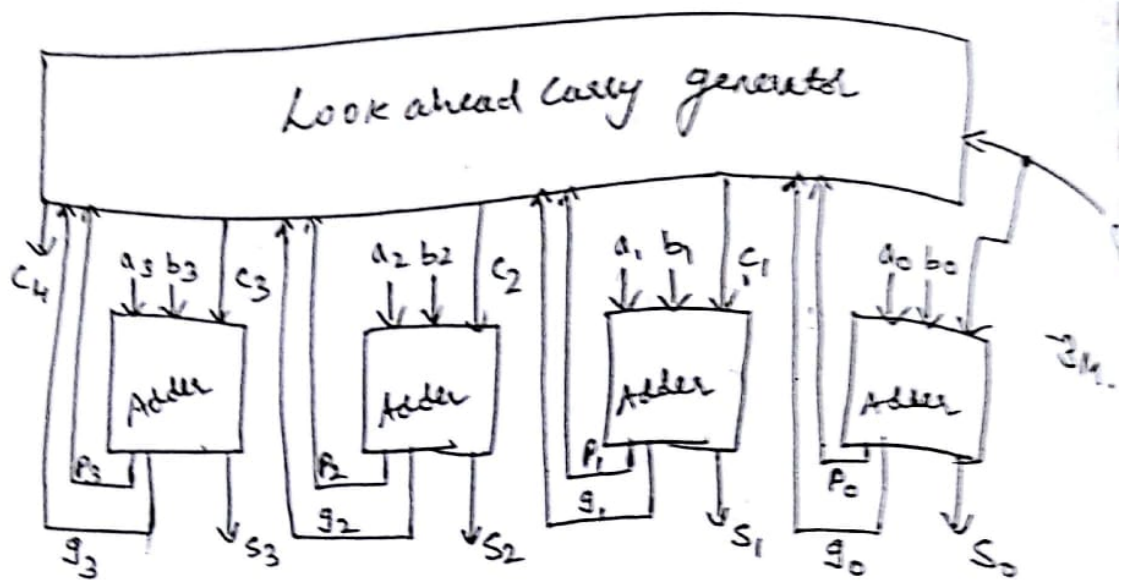
$$S_1 = \bar{B}_3 B_1 + B_3 B_2 B_1$$

$B_3 B_2$	$B_1 B_0$	00	01	11	10
00					
01					
11		1	1	1	1
10				1	1

$$S_2 = \bar{B}_3 B_2 + B_2 B_1$$



2) Look ahead carry scheme.



$$S_i = a_i \oplus b_i \oplus C_i \quad \text{--- (1)}$$

$$C_{i+1} = a_i b_i + C_i (a_i + b_i) \quad \text{--- (2)}$$

$$C_{i+1} = a_i b_i + C_i (a_i + b_i) \quad \text{--- (a)}$$

$$g_i = a_i b_i \quad \text{--- (b)}$$

$$P_i = a_i + b_i - C_i \quad \text{--- (c)}$$

$$C_{i+1} = g_i + P_i C_i \quad \text{--- (d)}$$

$$C_1 = g_0 + P_0 C_0 \quad \text{--- (e)}$$

$$C_2 = g_1 + P_1 C_1 \quad \text{--- (f)}$$

$$C_3 = g_2 + P_2 g_1 + P_2 P_1 g_0 + P_2 P_1 P_0 C_0 \quad \text{--- (g)}$$

$$C_4 = g_3 + P_3 g_2 + P_3 P_2 g_1 + P_3 P_2 P_1 g_0 + P_3 P_2 P_1 P_0 C_0 \quad \text{--- (h)}$$

Adv:- 2M.

(3)

Page No. _____
Date / / 20

$$f_1 = (\bar{y} + x)z$$

$$f_2 = (\bar{y}\bar{z} + \bar{x}y + y\bar{z})$$

$$f_1 = z\bar{y} + xz$$

$$= (x + \bar{x})\bar{y}z + x(y + \bar{y})z$$

$$= \underline{x\bar{y}z} + \bar{x}\bar{y}z + x\underline{y}z + x\bar{y}z \quad - 3M -$$

$$= \bar{x}\bar{y}z + x\bar{y}z + x\underline{y}z$$

$$= \Sigma(1, 5, 7)$$

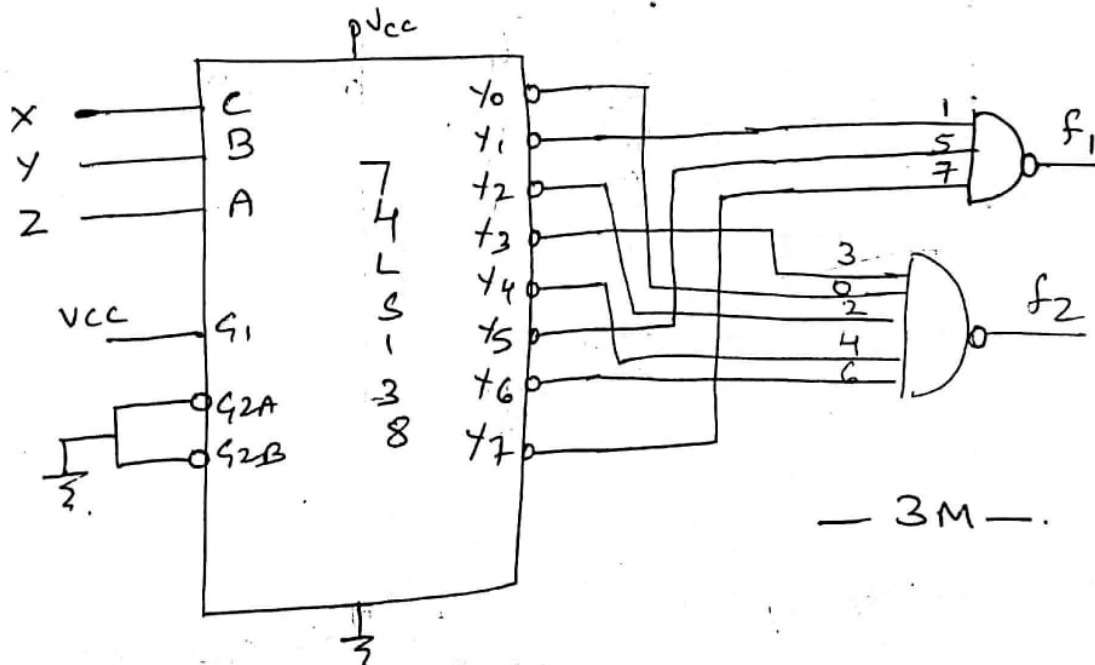
$$f_2 = \bar{y}\bar{z}(x + \bar{x}) + \bar{x}y(z + \bar{z}) + y\bar{z}(x + \bar{x})$$

$$= x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + \bar{x}y\underline{z} + \bar{x}y\bar{z} + x\underline{y}\bar{z} + \underline{x}y\bar{z}$$

$$= \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\underline{z} + x\underline{y}\bar{z} + x\underline{y}z$$

$$= \Sigma(0, 2, 3, 4, 6)$$

- 4M -



- 3M -

H) $f(a, b, c, d) = \Sigma m(1, 5, 7, 8, 9, 13, 14)$
using suitable multiplexer with a, b & c as control inputs.

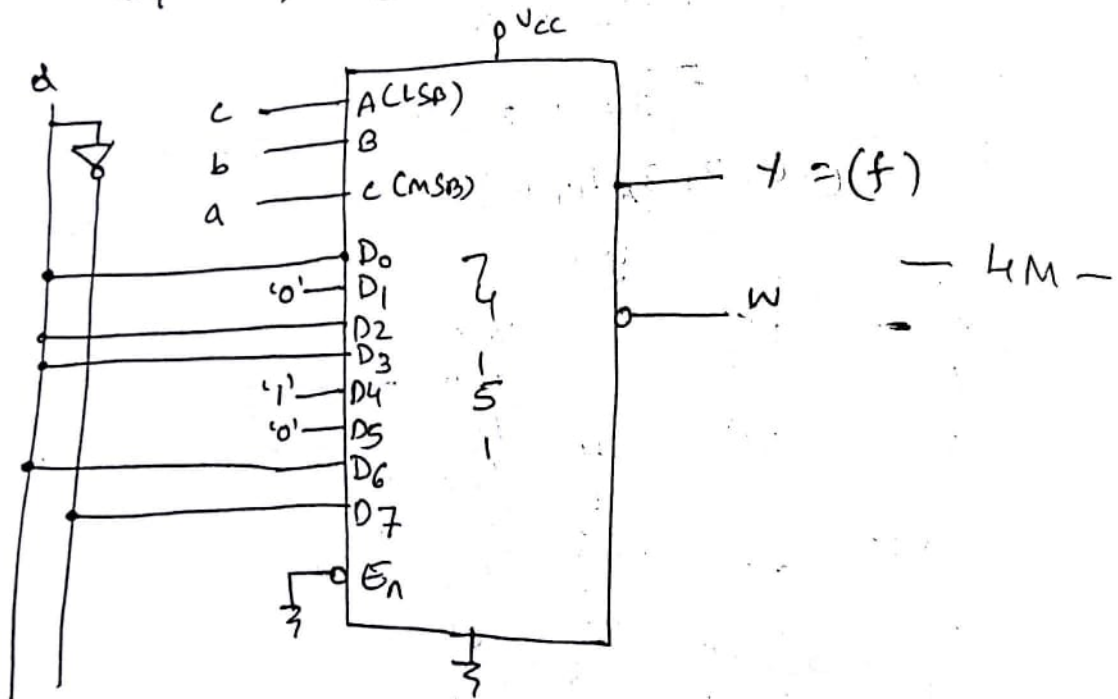
prepared by

Soln.

	$c=0$		$c=1$	
ab	00	01	11	10
00	D_0 1	D_1 0	D_2 0	D_3 0
01	D_4 1	D_5 1	D_6 0	D_7 1
11	D_8 1	D_9 1	D_{10} 0	D_{11} 0
10	D_{12} 1	D_{13} 1	D_{14} 0	D_{15} 0

a b & c as
Select inputs
we have to
derive 'd'
input
Variable

$D_0 = d$, $D_1 = 0$, $D_2 = d$, $D_3 = d$
 $D_4 = 1$, $D_5 = 0$, $D_6 = d$, $D_7 = \bar{d}$ - 2M.



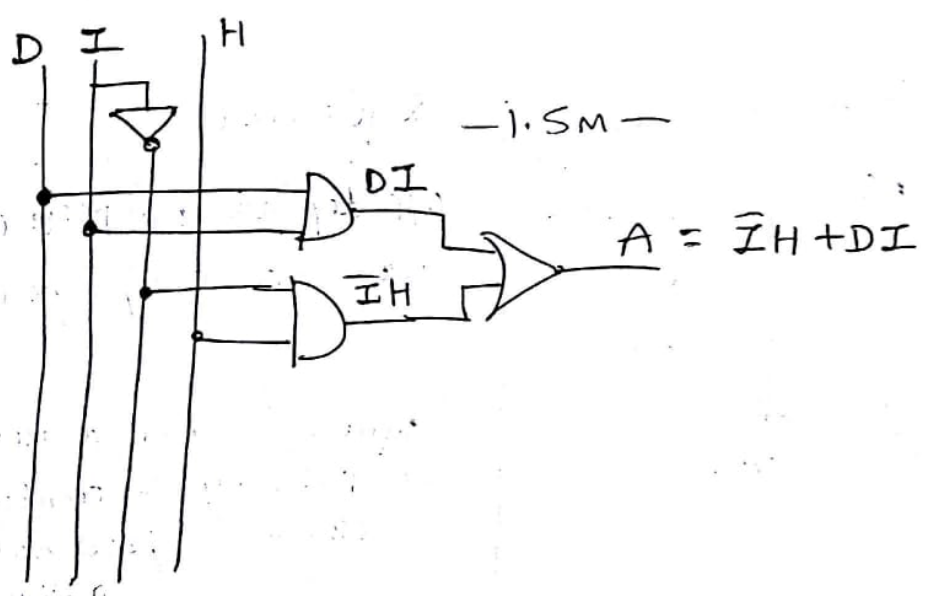
5a) 3 inputs as D, I, H
 ↓ ↓ ↓
 Door Ignition Head lights
 & o/p as
 A → Alarm.

④

D	I	H	A
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

D	I _H	00	01	11	10
0		0	1	0	0
1		0	1	1	1

$A = \bar{I}H + DI$
-1.5M-



5b) SR-FF

SR	Q	Q ⁺
0	0	0
0	0	1
0	1	0
0	1	0
1	0	1
1	0	1
1	1	1
1	1	1

SR	00	01	11	10
0	0	0	-	1
1	1	0	-	1

$Q^+ = S + Q\bar{R}$

-1.5M-

Prepared by
Sudhakar N

JK - FF

J	K	Q	Q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK	00	01	11	10
Q	0	0	1	1
1	1	0	0	1

$$Q^+ = J\bar{Q} + \bar{K}Q$$

- 1.5M -

Combinational

Sequential

- 2M -

- 1) O/p depends only on the present value of input
- 2) No memory
No feedback

O/p depends on present & past value of input

Sequential Ckts always have a memory associated with them. (feedback is present)

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Department of Electronics and Communication Engineering

Semester: III

Test-II

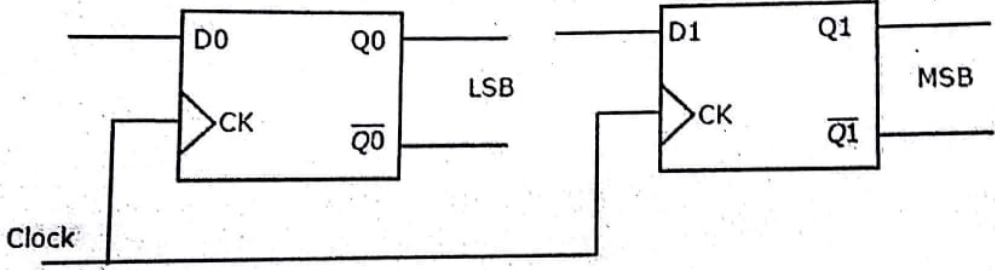
Course Name: Analysis and Design of Digital Circuits
 Time : 2 hours

Course Code: 12EC3
 Date: 6/10/2016
 Maximum Marks: 65

Instructions to candidates:

1. Answer all the questions from PART-A. PART-A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART-A.

PART-A

Sl.No	QUESTION	M	BT	CO
1	Two D flip-flops are connected as a synchronous counter as shown in Fig.1. The counter goes through the Q_1Q_0 sequence 00 → 01 → 11 → 10 → 00 → The inputs D_0 and D_1 respectively are _____ & _____.  Fig.1	2	4	4
2	A K-bit Johnson counter produces _____ number of timing signals.	1	1	3
3	Define 'set up' and 'hold time' with respect to edge triggered flip flops.	1	1	3
4	List any two methods to eliminate race around condition in a JK FF.	1	2	1
5	_____ number of flip flops are complemented in a 10 bit binary ripple counter after the following count 1001100111.	1	4	2
6	Determine f_{max} for a 4-bit synchronous counter if, t_{pd} for each flip flop is 50ns and t_{pd} for each and gate is 20ns. Compare this with f_{max} for a mod-16 ripple counter.	2	3	4

7

In a JK flip-flop shown in Fig.2, $J=Q'$ and $K=1$. Assume the flip flop was initially cleared and then clocked for 6 pulses. The sequence at the Q output will be _____.

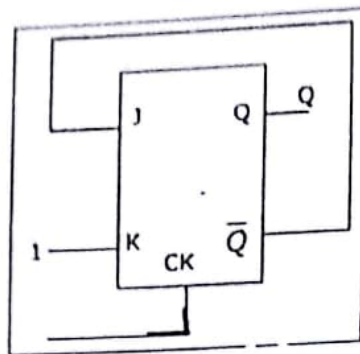


Fig.2

8

Draw the state diagram for the logic circuit shown in Fig.3.

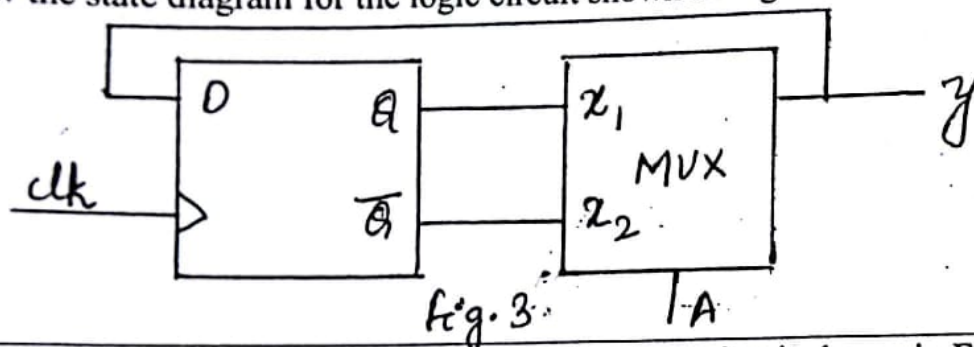


Fig.3

9

Draw the timing diagram of output (Y) in the given circuit shown in Fig.4

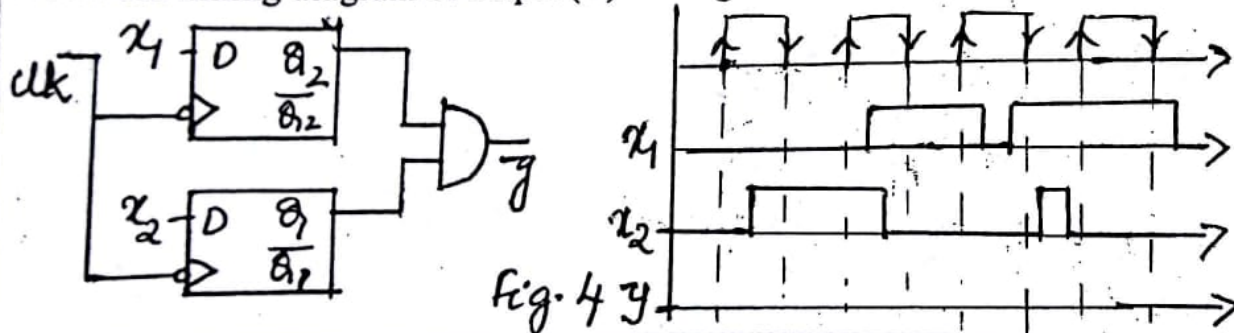


Fig.4

10

Using IC-74193, design adown counter which counts from 1101 to 0011(13-3).

11

A sequential circuit using a D flip flop and logic gates is shown in Fig.5. X and Y are the inputs and Z is output. The circuit is _____.

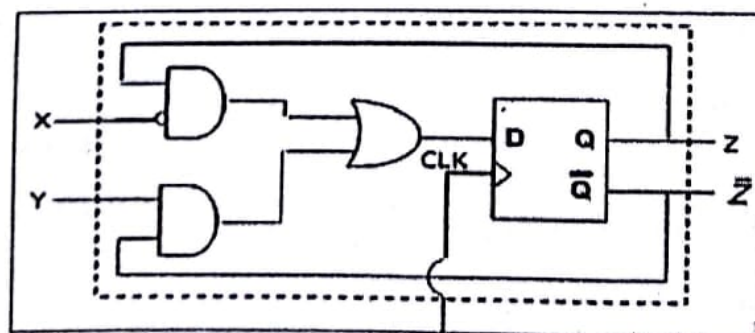
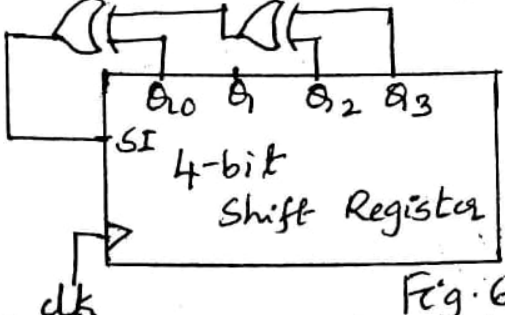
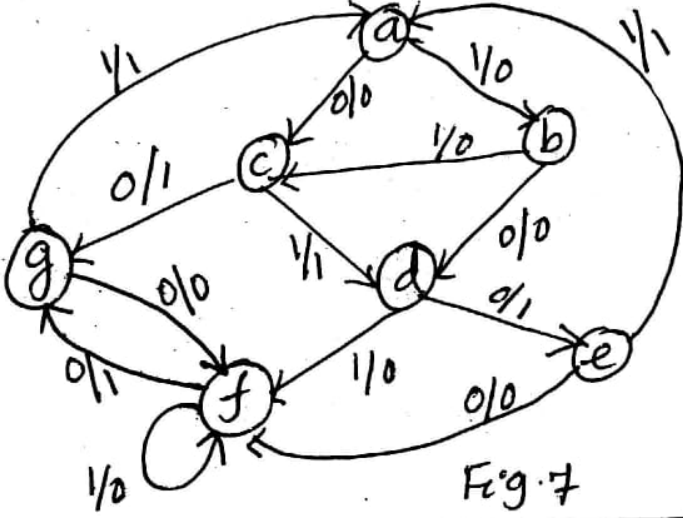


Fig.5

PART-B

Sl.No	QUESTION	M	BT	C
1 a)	A 4-bit shift register which shifts 1-bit to the right at every clock pulse, is initialized the pattern (1000) for ($Q_0Q_1Q_2Q_3$). The D input is derived from Q_0, Q_2 and Q_3 through XOR gates as shown in Fig.6.  Fig. 6 i) Write the 4-bit pattern ($Q_0 Q_1 Q_2 Q_3$) after each clock pulse till pattern (1000) reappears on ($Q_0 Q_1 Q_2 Q_3$). ii) To what value should the shift register be initialized so that the pattern (1001) occurs after the first clock pulse.	5	4	
1 b)	Design a positive edge triggered D-FF and explain its operation.	5	3	3
2	Design a clocked sequential circuit for the state diagram shown in Fig.7, using T-FFs. Also draw the reduced state diagram.	10	5	4
	 Fig. 7			
3 a)	Design an Asynchronous BCD counter using JK-FFs. Assume Preset and Clear inputs are disabled permanently.	6	3	1

3 b)	For the given set of operations, design a 4-bit universal shift register.	4	2										
	<table><tr><th>Mode</th><th>operation</th></tr><tr><td>00</td><td>Hold</td></tr><tr><td>01</td><td>Shift Left</td></tr><tr><td>10</td><td>Shift Right</td></tr><tr><td>11</td><td>Complement the content</td></tr></table>	Mode	operation	00	Hold	01	Shift Left	10	Shift Right	11	Complement the content		
Mode	operation												
00	Hold												
01	Shift Left												
10	Shift Right												
11	Complement the content												
4 a)	Design a sequential circuit with JK FF to satisfy the following state equations: $A(t+1) = A'B'CD + A'B'C + AC'D'$ $B(t+1) = A'C + CD' + A'BC'$ $C(t+1) = B$ $D(t+1) = D'$	5	4										
4 b)	Construct a Johnson counter with 10 timing signals. Compare Johnson and Ring counters.	5	3										
5 a)	Describe the working of a JK master slave flip-flop with timing diagrams.	5	2										
5 b)	Draw the block diagram of a 4-bit serial adder using a 2-bit counter which outputs signal K after 11 state is reached. When start signal N is received, the register should be loaded. Assume that N=1 until addition is complete. When the addition is complete, the control network should go to a stop state and remain there until N is changed back to 0.	5	4										

BT-Blooms Taxonomy,CO-Course Outcomes,M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5
	Quiz	Max Marks	4	3	4	4	2	2	3	6	2
	Test	Max Marks	15	10	10	15	-	9	16	15	10

(1)

RUCG

Dept of ECE

Scheme - II

Test - II

prepared by
12c. sanyal
associated prof

Semester - 3
Time - 2 hours

Course Code: 12EC34

Date: 6/10/2016

Part - A

- 1) The inputs D_0 & D_1 respectively are $\bar{Q}_1$ & Q_0 (2M)
- 2) A K bit Johnson counter produces $\frac{2K}{2}$ number of timing signals. (1M)

- 3) { 1M.
 Setup time :- The time it must be fixed before the clock edge is called the setup time
 Hold time :- The time it must be fixed after the clock edge is called the hold time

NOTE: A edge triggered latch (FF) ideally samples the data line instantaneously on one of the edges of the clock. However, nothing is truly instantaneous, so the data must be valid for some finite amount of time around the clock edge.

- 4) Two methods to eliminate race around condition are (1M)
1. Master slave flip flop
 2. positive & negative edge triggered flip flops.

- 5) Four number of flip flops are complemented. (1M)

following count 1001100111.

NOTE: we simply need to figure out what the next count in the sequence is & find how many bits have changed.

- a) Add 1 to the value & we have (1001101000).
So the first four bits toggle.

- ⑥ Solution: For a synchronous counter the total delay that must be allowed b/w input clock pulses is equal to flip-flop t_{pd} + no. of gate t_{pd} . Thus $T_{clock} \geq 50 + 20 = 70 \text{ ns}$, & so

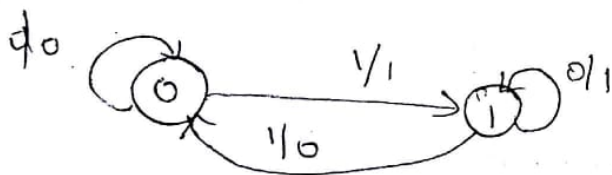
the counter has
 $f_{max} = \frac{1}{70 \text{ ns}} = 14.3 \text{ MHz}$. — (1M)

WKT MOD-16 ripple counter used four flip-flops with flip-flop $t_{pd} = 50 \text{ ns}$, the f_{max} for ripple counter can be given as

$$f_{max}(\text{ripple}) = \frac{1}{4 \times 50 \text{ ns}} = 5 \text{ MHz}. \text{ — (1M)}$$

- ⑦ The sequence at the Q₂ o/p will be 01010101 — (2M)

8)



(2M) when $y=0$

$$A=0 \quad x_1=0$$

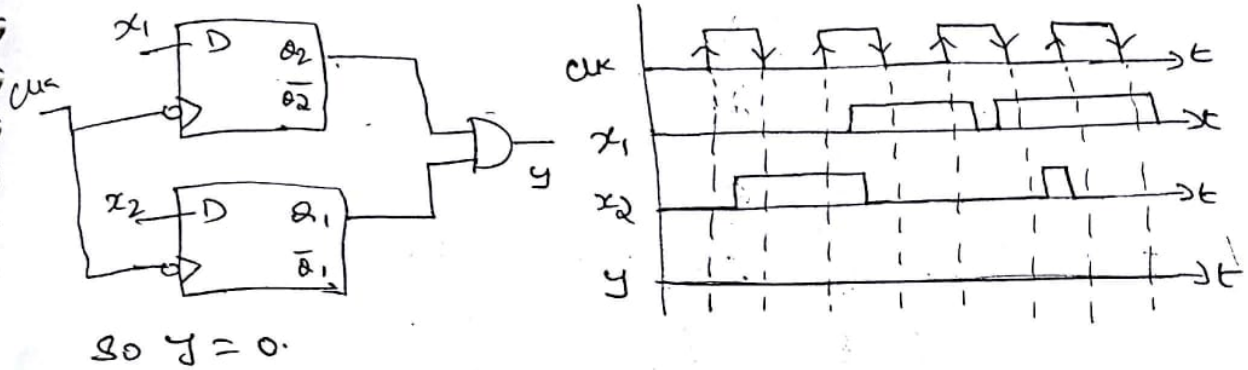
$$A=1 \quad x_1=1$$

when $y=1$

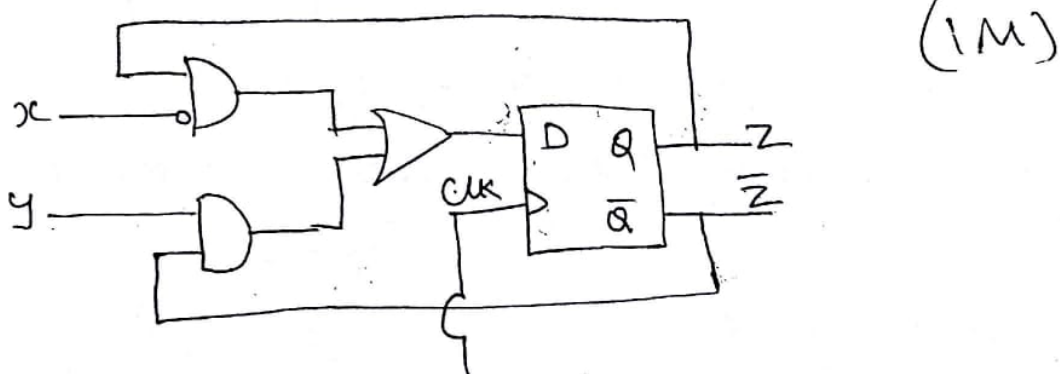
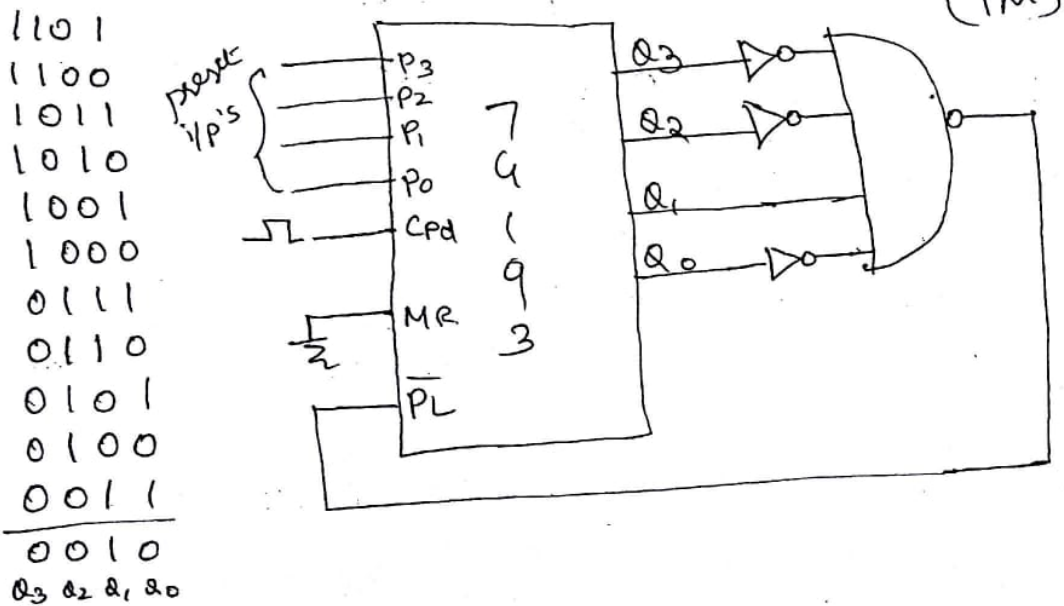
$$A=0 \quad y=1$$

$$A=1 \quad y=0$$

9) Draw Timing diagram of o/p y in the given.
 Ckt Shown in fig. below. —(1M)

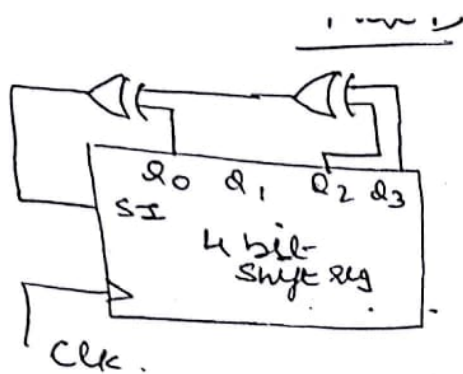


10)



Ans. J-K flipflop with inputs $x = K$ & $y = J$

(a)



u) write the 4-bit values (Q_0, Q_1, Q_2, Q_3)

— SM

Shift register register $Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$

$SI/D = Q_0 \oplus Q_2 \oplus Q_3$ (D can be derived)

D	clk	Q_0	Q_1	Q_2	Q_3
—	—	1	0	0	0
1	1	1	1	0	0
1	2	1	1	1	0
0	3	0	1	1	1
0	4	0	0	1	1
0	5	0	0	0	1
1	6	1	0	0	0

— initial content

odd no. of one's

→ Sequence repeats.

b)

1001 — State (After clock pulse)

After 1st clock pulse Q_0, Q_1, Q_2, Q_3

$$D \oplus Q_0 \oplus Q_2 = 1001$$

$$Q_0 = 0, Q_1 = 0, Q_2 = 1$$

$$D = 1 = Q_0 \oplus Q_2 \oplus Q_3, D = 1 = 0 \oplus 1 \oplus Q_3$$

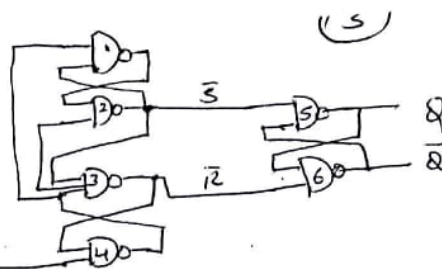
$$Q_3 = 0$$

If we going for initial 0010 After first clock pulse 1001

1b) the edge triggered DFF

Diagram - 2M

Explanation - 3M.



Design closed sequential ckt using TFF's

Design - 5M

Logic diagram - 2M.

Reduced State Diagram - 2M

Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0

PS	NS		O/P	
	x=0	x=1	x=0	x=1
a	c	b	0	0
b	d	c	0	0
c	e	d	1	1
d	e	d	1	0
e	d	a	0	1

$$y = BC + A(C \oplus x)$$

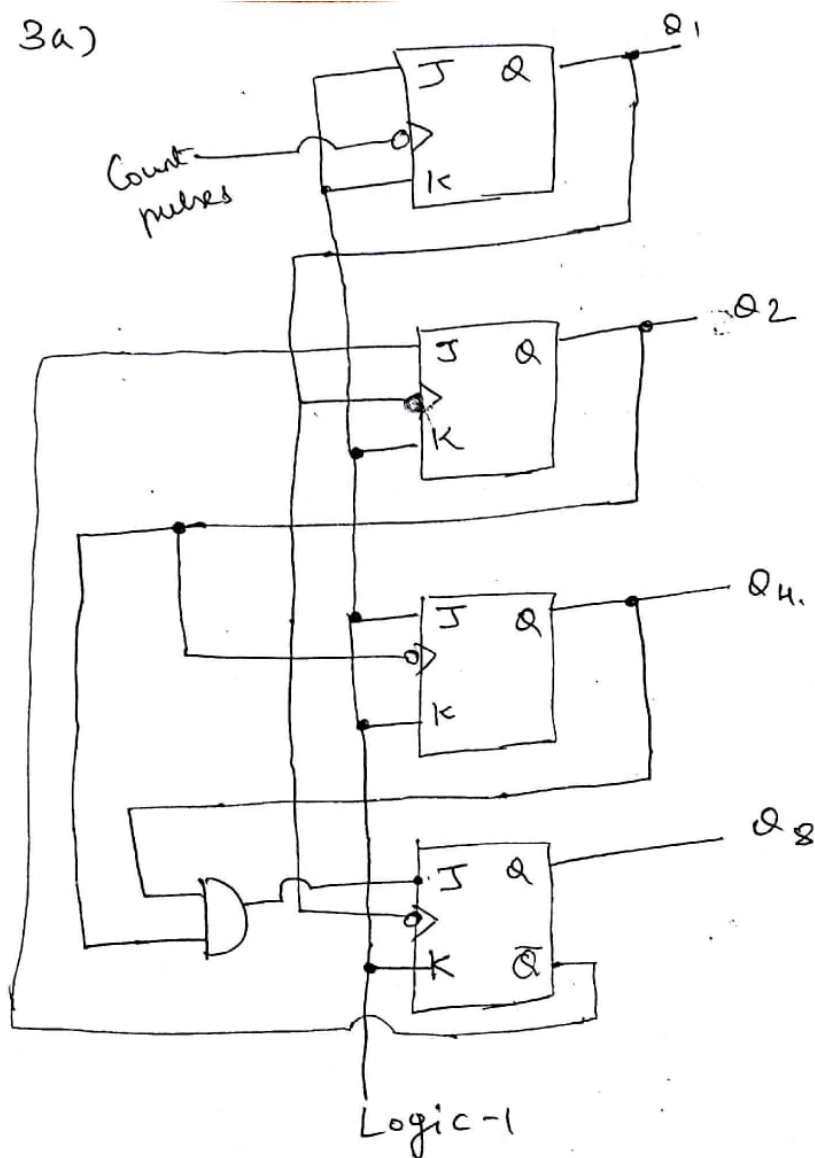
$$T_A = CB + \bar{x}B + ACx$$

$$T_B = \bar{A}C + \bar{A}\bar{x}$$

$$T_C = \bar{A}x + xB + \bar{C}\bar{x}$$

P S				NS			excitation			O/P
A	B	C	x	A ⁺	B ⁺	C ⁺	T _A	T _B	T _C	y
0	0	1	0	0	1	1	0	1	0	0
0	0	1	1	0	1	0	0	1	1	0
0	1	0	0	1	0	0	1	1	0	0
0	1	0	1	0	1	1	0	0	1	0
0	1	1	0	1	0	1	1	1	0	1
0	1	1	1	1	0	0	1	1	1	1
1	0	0	0	1	0	1	0	0	1	1
1	0	0	1	1	0	0	0	0	0	0
1	0	1	0	1	0	0	0	0	1	0
1	0	1	1	0	0	1	1	0	0	1

3a)



Logic diagram - 2M
Explanation - 4M

3b) universal shift register
Designing - 4M.

40) i) $A(t+1) = \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C + A\bar{C}\bar{D} + A\bar{C}\bar{D}$ (1)
 $J = \bar{B}C$; $K = C$ — 1M

ii) $B(t+1) = \bar{A}C + C\bar{D} + \bar{A}B\bar{C}$ — 1M.
 $J = \bar{A}C + C\bar{D}$; $K = A\bar{C} + A\bar{D}$

iii) $C(t+1) = B$
 $J = B$; $\bar{K} = B$ but $K = \bar{B}$ } — 1M-
 iv) $D(t+1) = \bar{D}$
 $J = 1$ & $K = 1$.

Designing a Sequential Ckt — 2M —

45) (b) Compare Johnson & Ring Counter — 2M

Ring Counter

Johnson Counter

- ① Ring counter consisting of n FF's has only n states in its counting sequence
 (4 FF's — 4 states)
 [Single bit is circulated]

A pattern of bits equal in length to twice the length of the shift reg thus circulates indefinitely

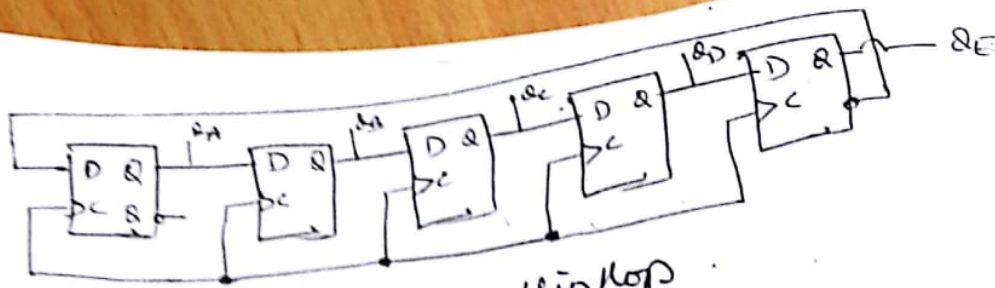
where the o/p from the last stage is inverted & fed back as input to the first stage

- ② A ring counter is a shift register with the o/p of the last one connected to the input of the first, i.e., in a ring.

a) Construct a Johnson Counter
 (5 DFF's are required)

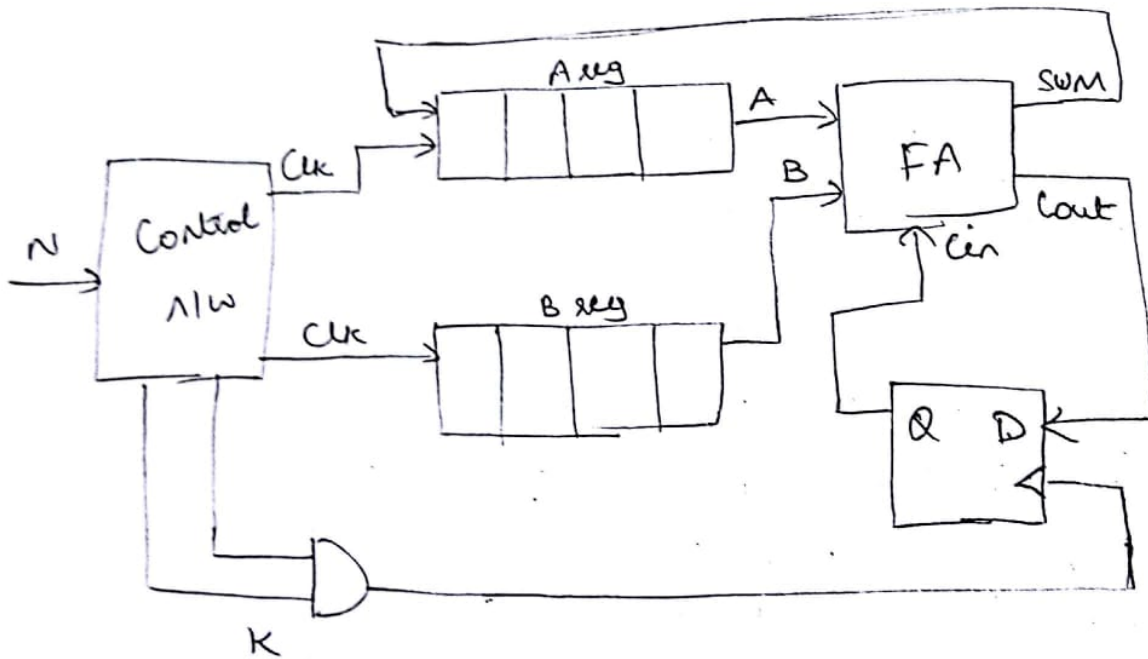
— 3M —

(Diagram + TT)
 (2M) + (1M)



- 5a) JK master slave flipflop
 Logic diagram — 2M
 Explanation — 3M.

- 5b) Block diagram of a 4-bit serial adder using 2-bit counter — 5M.



R.V.COLLEGE OF ENGINEERING, Bengaluru-560059
(Autonomous institution affiliated to VTU, Belagavi)
Department of Electronics and Communication Engineering

Semester:III
Test-III

Course Name: Analysis and Design of Digital Circuits
Time :2hours

Course Code: 12EC34
Date: 2/11/2016
Maximum Marks:65

Instructions to candidates:

1. Answer all the questions from PART-A. PART-A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART-B.

PART-A

Sl.No	QUESTION	M	BT	CO
1	The fastest Digital IC logic family is _____.	1	1	2
2	A logic gate in a particular logic family has following specifications: $V_{th} = 2$ volts, $V_{OH} = 4$ volts, $V_{IH} = 3$ volts, $V_{OL} = 1$ volt, and $V_{IL} = 1.5$ volts. Its noise margin is _____.	2	3	3
3	Define Propagation delay and Figure of merit.	1	1	2
4	The logic family which has the lowest power dissipation and lowest speed is _____.	1	1	1
5	If the output high current of a gate is 500mA and its input high current is 50mA, the fan out of the gate is _____.	1	3	2
6	What is the function performed by a parallel adder with inputs A and B with all bits of B made as logic 1 with $C_{in}=1$?	1	3	4
7	The status register condition after the operation of unsigned subtraction (A-B), if $A > B$ is _____.	1	2	3
8	Realize AND gate using Transmission gate.	2	4	3
9	A processor unit employs a scratchpad memory and it consists of 128 registers of eight bits each. i) How many lines are needed for address? ii) How many lines are there for the input data?	2	2	1
10	In the status register of 4 bit ALU, the result of a signed operation sets the V bit, if the result exceeds _____.	1	3	4
11	Design an inverter(NOT) gate using nMOS logic.	2	4	2

PART-B

PART-B

Sl.No	QUESTION	M	BT	CO																																																						
1.	With a neat diagram, explain the operation of NOR/OR gate using Emitter Coupled Logic (ECL). Also mention the advantages and disadvantages of ECL.	10	2	2																																																						
2	Design an arithmetic and logic circuit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.	10	5	4																																																						
	<table border="1" style="width: 100%; text-align: center;"> <thead> <tr> <th colspan="5">Arithmetic Operations</th> <th colspan="4">Logical Operations</th> </tr> <tr> <th>S_2</th> <th>S_1</th> <th>S_0</th> <th>$C_{in}=0$</th> <th>$C_{in}=1$</th> <th>S_2</th> <th>S_1</th> <th>S_0</th> <th>$C_{in}=X$</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0</td> <td>0</td> <td>$F=A$</td> <td>$F=A+1$</td> <td>1</td> <td>0</td> <td>0</td> <td>OR</td> </tr> <tr> <td>0</td> <td>0</td> <td>1</td> <td>$F=A+B$</td> <td>$F=A+B+1$</td> <td>1</td> <td>0</td> <td>1</td> <td>XOR</td> </tr> <tr> <td>0</td> <td>1</td> <td>0</td> <td>$F=A-B-1$</td> <td>$F=A-B$</td> <td>1</td> <td>1</td> <td>0</td> <td>AND</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> <td>$F=A-1$</td> <td>$F=A$</td> <td>1</td> <td>1</td> <td>1</td> <td>Complement</td> </tr> </tbody> </table>	Arithmetic Operations					Logical Operations				S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$	0	0	0	$F=A$	$F=A+1$	1	0	0	OR	0	0	1	$F=A+B$	$F=A+B+1$	1	0	1	XOR	0	1	0	$F=A-B-1$	$F=A-B$	1	1	0	AND	0	1	1	$F=A-1$	$F=A$	1	1	1	Complement			
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0	1	1	$F=A-1$	$F=A$	1	1	1	Complement																																																		
3 a)	Draw and explain the working of NAND and NOR gates using CMOS logic. Also mention the advantages of CMOS logic.	6	3	3																																																						
3 b)	Write a short note on status register of a processor.	4	1	1																																																						
4 a)	A 74LS00 NAND gate output is driving three 74LS00 NAND gates and two 74AS00 NAND gates. Determine if there is a loading problem. The specifications for the same are:	4	3	2																																																						
	<table border="1" style="width: 100%; text-align: center;"> <thead> <tr> <th></th> <th>74LS00</th> <th>74AS00</th> </tr> </thead> <tbody> <tr> <td>I_{OH}</td> <td>-0.4mA</td> <td>-2mA</td> </tr> <tr> <td>I_{IH}</td> <td>20μA</td> <td>20μA</td> </tr> <tr> <td>I_{OL}</td> <td>8mA</td> <td>20mA</td> </tr> <tr> <td>I_{IL}</td> <td>-0.4mA</td> <td>-0.5mA</td> </tr> </tbody> </table>		74LS00	74AS00	I_{OH}	-0.4mA	-2mA	I_{IH}	20 μ A	20 μ A	I_{OL}	8mA	20mA	I_{IL}	-0.4mA	-0.5mA																																										
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4 b)	What is a scratch pad memory? Explain the organization of a processor unit using scratch pad memory with the help of a neat diagram.	6	2	1																																																						
5	For the following set of micro operations, write control word register content. i) $R1 \leftarrow Shr R2$ ii) $R2 \leftarrow Shl R6$ iii) $R1 \leftarrow Crc R2$ iv) $R1 \leftarrow R5 - R7$ v) $R3 \leftarrow R5 xor R5$.	10	4	3																																																						

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	3	5	5	2	3	3	5	4	-	-
	Test	Max Marks	8	16	16	10	4	14	12	10	10	-

RNCE, Dept of ECE
Scheme

Sub: ADDC.

Test: 3

Course code: 12EC34

Max marks: 65

Prepared by
Dr. Shylashree N.

Quiz - Part-A

ECL - 1M -

0.5 Volts - 2M -

- a) Propagation delay of a gate is basically the time interval between the application of an input pulse & the occurrence of the resulting O/P pulse. - 1M -
- b) Figure of merit :- It is defined as product of propagation delay & Average power dissipation
- ∴ Speed power product = $\frac{\text{propagation delay (seconds)}}{\text{(J)}} \times \text{power dissipation (Joules/seconds)}$

CMOS family - 1M -

$$I_{OH} = 500 \mu A \quad \text{fanout} = \frac{I_{OH}}{I_{IH}} = \frac{500}{50} = 10 \quad - 1M -$$
$$I_{IH} = 50 \mu A$$

All bits of B made as logic '1' with $C_{in} = 1$

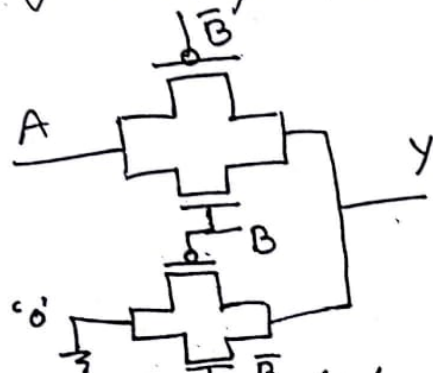
$Y = A$ operation is transfer A - 1M -

Unsigned subtraction $(A - B)$ if $A > B$ is - 1M -

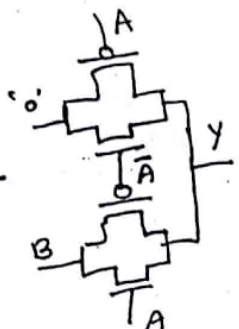
Status register condition is $Z = 0$ & $C = 1$

8)

AND gate using Transmission gate



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



-2M-

9)

- i) 7 lines are needed for address
- ii) 8 lines are there for the input data

-2M-

10)

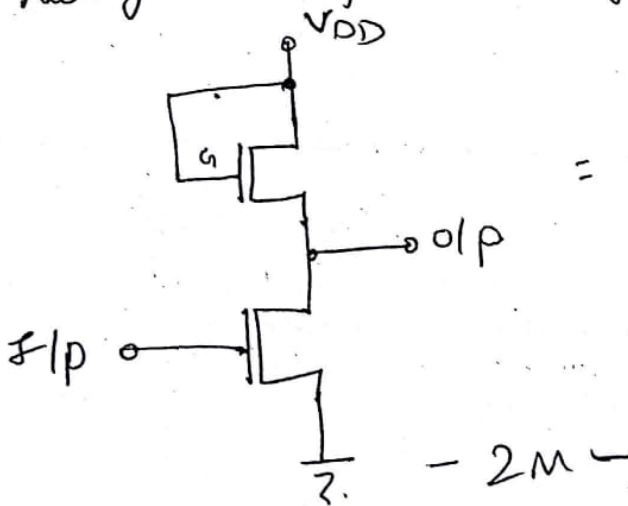
Numeric range of o/p.

i.e. If result is greater than & equal to 2^4
 result ≥ 16

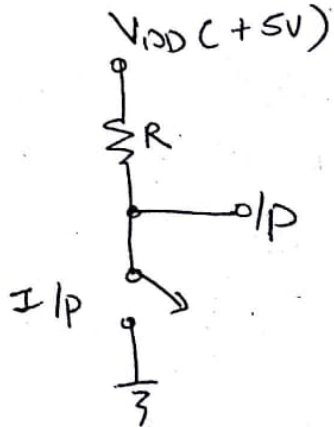
-1M-

11)

NOT gate using CMOS logic.

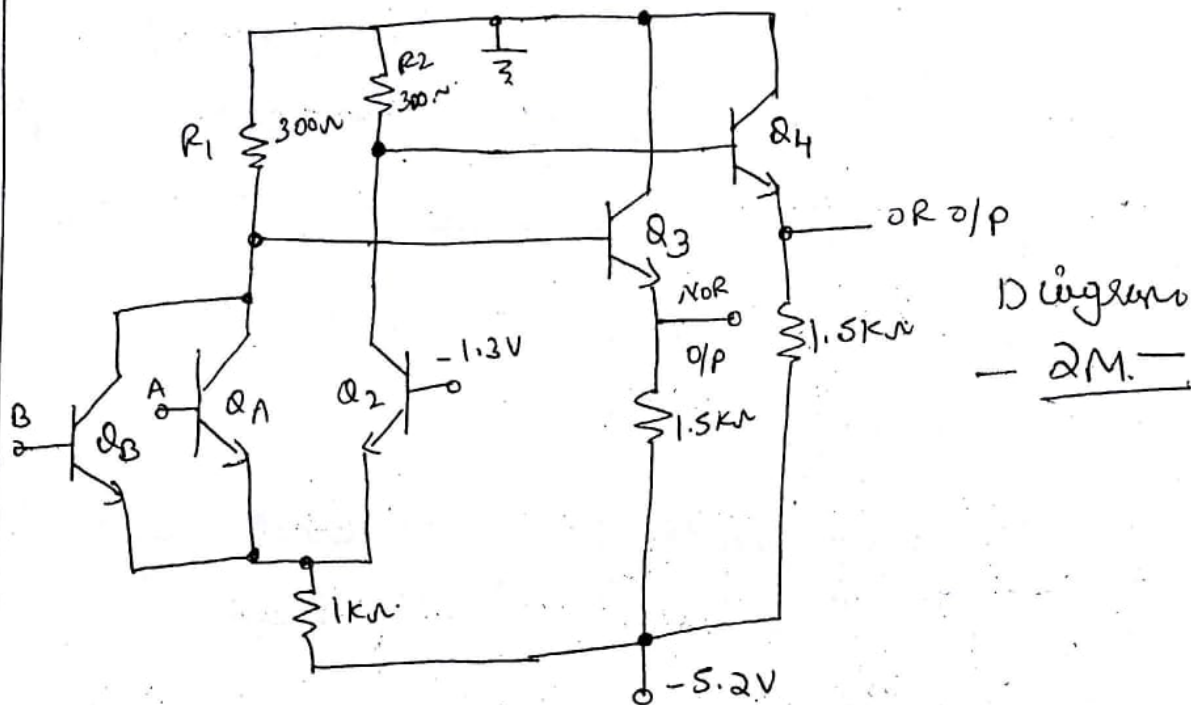


=



-2M-

PART-B



operation — 4M —

Advantages — 2M —

- ① It is a non-saturated logic, in the sense that the transistors are not allowed to go in to saturation. So, storage time delays are eliminated & therefore, the speed of operation is increased.
- 2) currents are kept high, & the o/p impedance is so low that circuit & stray capacitance can be quickly charged & discharged.
- 3) The limited V_{th} swing.

Disadvantages — 2M —

- 1) High cost
- 2) Low noise margin
- 3) High power dissipation
- 4) problem of cooling.
- 5) Its negative supply voltage & logic levels are not compatible with other logic families.

2)

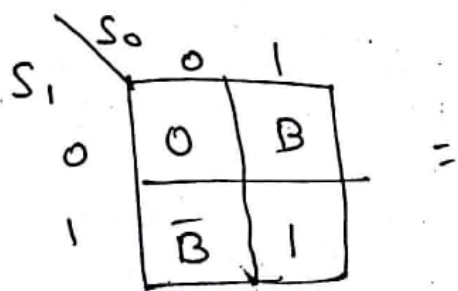
we can realize that by selecting proper B & C_{in} inputs we can implement basic arithmetic operations. There are four possible B inputs: True, Complement, all 1's & all 0's and two possible C_{in} inputs 0 & 1.

we can write truth table for B input with respect to S₀ & S₁ signals as follows.

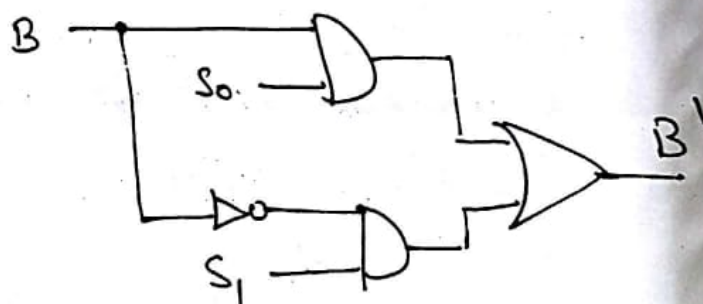
S ₁	S ₀	B'
0	0	0
0	1	B
1	0	$\bar{B}$
1	1	1

NOTE B' indicates Modified B input.

— 2M —

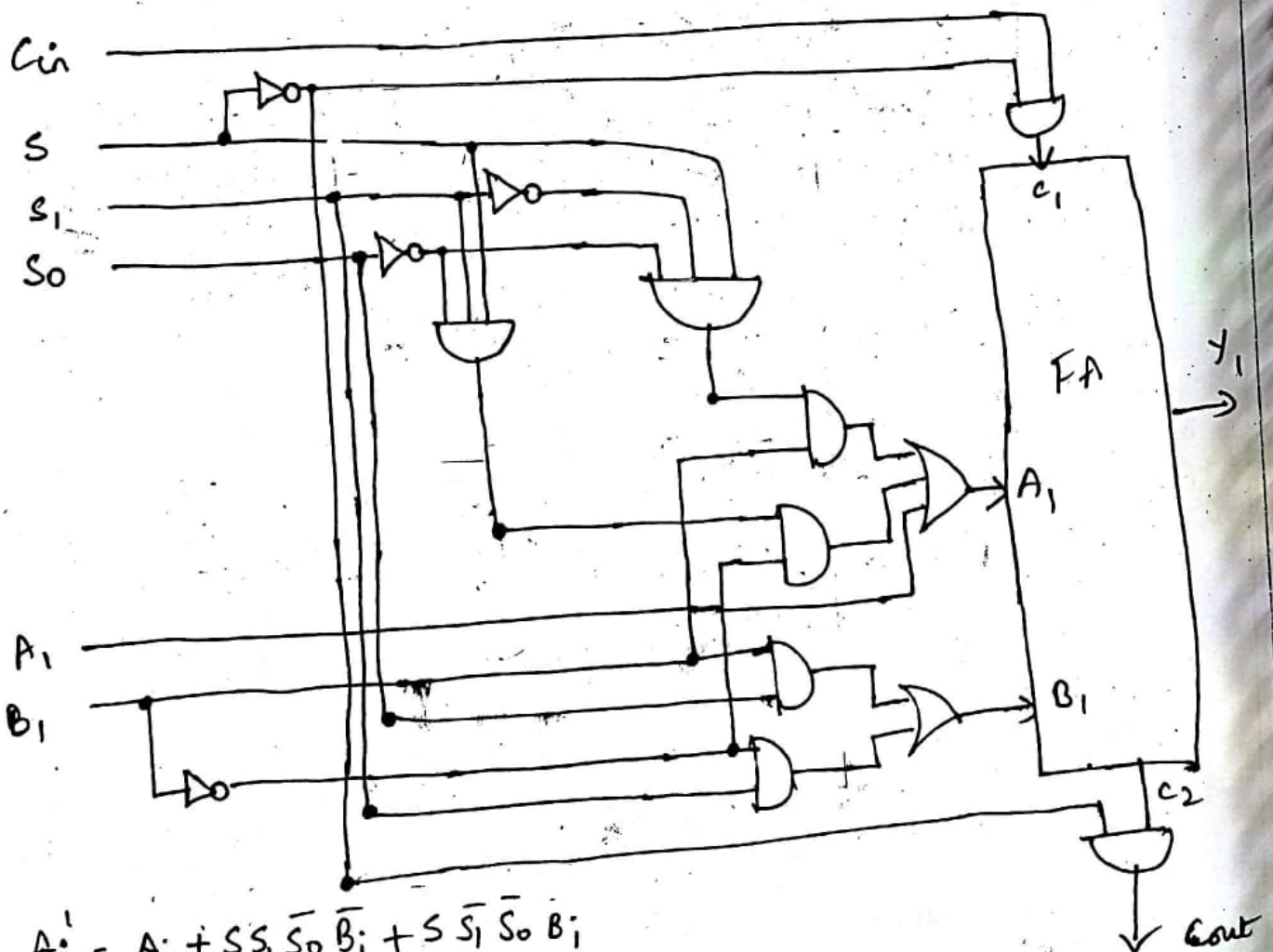
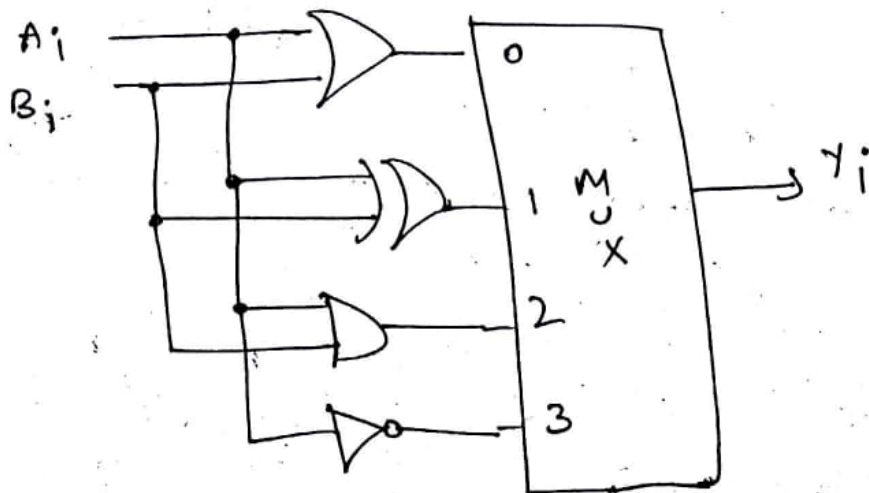


$$\therefore B' = S_1 \bar{B} + S_0 B$$



using the logic diagram of Modified B input we can draw the logic diagram of arithmetic, as shown below.

Design of logic circuit



$$A_1' = A_i + S S_1 \bar{S}_0 \bar{B}_i + S \bar{S}_1 \bar{S}_0 B_i$$

$$B_1' = S_1 \bar{B}_i + S_0 B_i$$

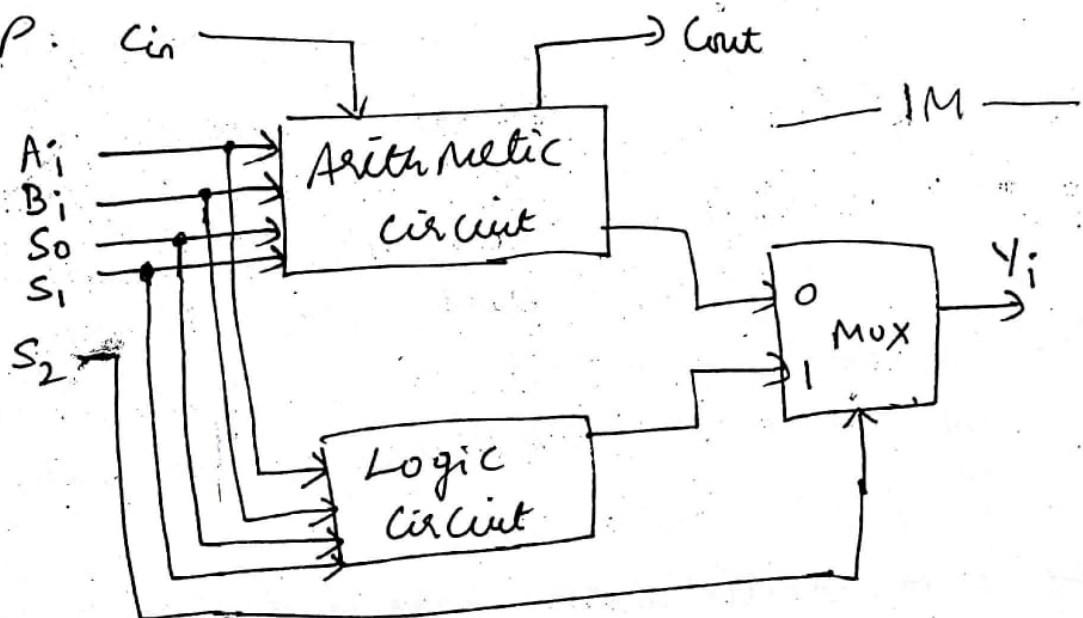
$$C_1' = \bar{S} C_i$$

— 4M —

+ Design

By combining with the help of MUX we can get the Arithmetic & logic ckt as shown in fig below.

when S_2 is 0, the o/p of arithmetic circuit is transferred as a final o/p; otherwise the o/p of logic ckt is transferred as a final o/p.



Combining arithmetic & logic ckt

a) Advantages of CMOS

- 1) It is also possible to fabricate enhancement mode p-channel & n-channel MOS devices on the same chip. Such devices are referred to as complementary MOSFET's & logic based on these devices is known as CMOS logic.

ii) The power dissipation is extremely small for CMOS & hence CMOS logic has become very popular.

NAND gate:- 2.5M -

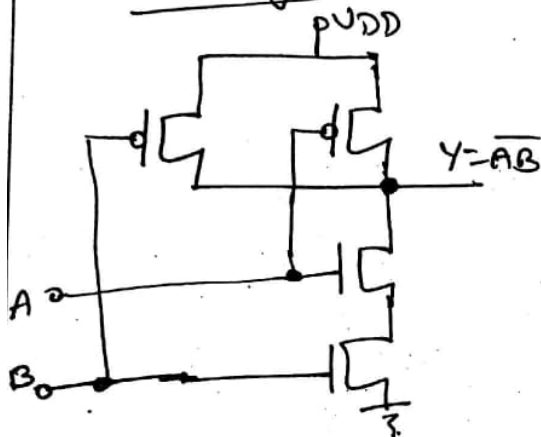


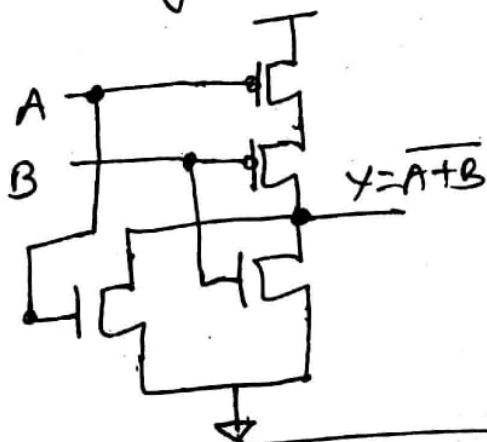
fig shows 2-input CMOS NAND gate. It consists of two series NMOS transistors b/w Y & GND & two like PMOS transistors b/w Y & VDD.

If either input A & B is '0' at least one of NMOS transistors will be OFF, breaking the path from Y to GND.

But at least one of the PMOS transistors will be ON, creating a path from Y to VDD. Hence the o/p Y will be '1'.

If both inputs are '1', both of the NMOS transistors will be ON & both of the PMOS transistors will be OFF. Hence, the o/p will be '0'.

NOR gate:- 2.5M -



If either input A & B is '1', at least one of the NMOS transistors will be ON & one of the PMOS transistors will be OFF, & the o/p Y is pulled down to '0'.

If both the inputs are '0', both of the pmos transistors will be ON & both of the nmos transistors will be OFF. Hence, the o/p will be '1'.

Status register of a processor — LM —

The Status register indicates certain conditions of the result such as.

- Result is Zero
- Result is Negative / positive
- Result with carry
- Result with overflow.

These conditions are indicated by certain bit conditions. Let us see bits in the status register.

C-bit : Carry bit: It is set when the o/p carry of ALU is 1; otherwise it is 0.

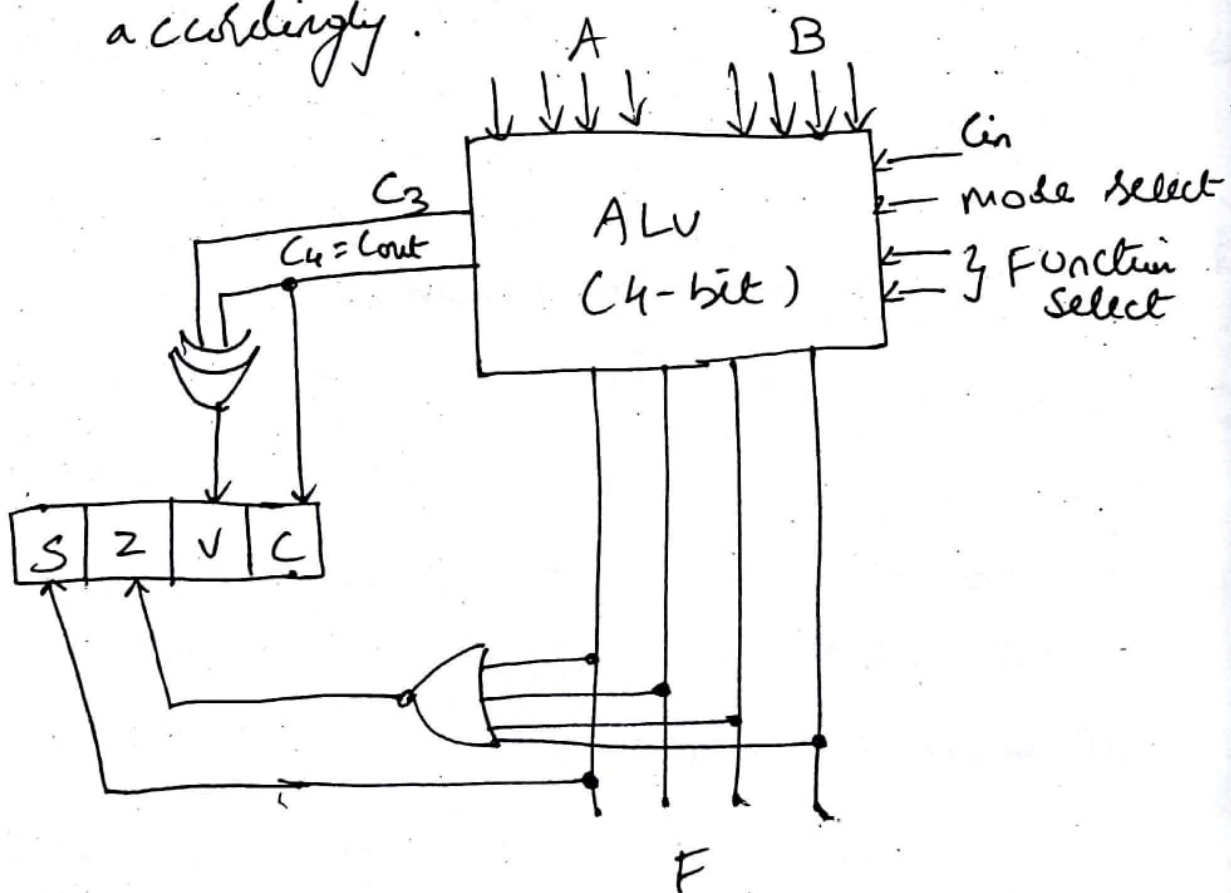
S-bit : Sign bit: It is set when MSB of the result is 1; otherwise it is Zero.

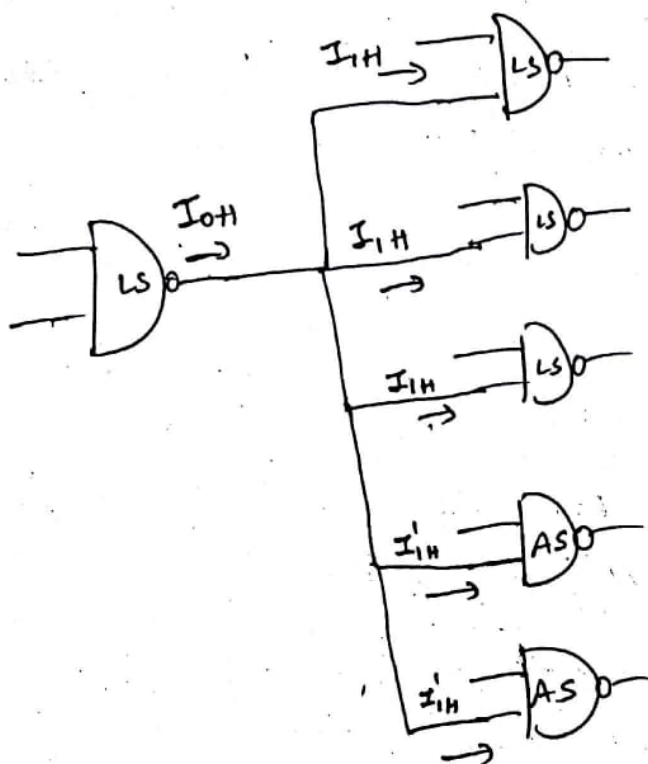
It indicates the sign of the result.

when S bit is 1, result is negative; otherwise result is true.

Z-bit : Zero bit: It is set when the result is Zero; otherwise it is Zero.

V-bit: overflow bit: In 2's complement arithmetic, most significant bit is used to represent sign & remaining bits are used to represent magnitude of a number. It is set if the result of a signed operation is too large to fit in the number of bits available to represent it. The overflow can be detected by exclusively-oring two most significant carry out signals. fig shows the logic circuit which detects above mentioned conditions & sets the flag bits accordingly.





$$f_{an out} = \frac{I_{OH}}{I_{IH}}$$

$$f_{an out} = \frac{I_{OL}}{I_{IL}}$$

→ So 74LS00 has fanout as 20

→ 74AS00 has fanout as 16

fan out for 74LS00

$$\frac{I_{OH}}{I_{IH}} = \frac{0.4 \text{ mA}}{20 \mu\text{A}} = 20$$

$$\frac{I_{OL}}{I_{IL}} = \frac{8}{0.4} = 20$$

fan out = 20

fan out for 74AS00

$$\frac{I_{OH}}{I_{IH}} = \frac{0.4 \text{ mA}}{20 \mu\text{A}} = 20$$

$$\frac{I_{OL}}{I_{IL}} = \frac{8}{0.5} = 16$$

fan out = 16 (consider the minimum)

No loading problem exists as only 5 gates are loaded.

Scratchpad memory: - |M-

processor unit consists of multiple registers, they can be arranged together in the form of a fast access memory called scratchpad memory

Scratchpad memory is different from the Computer main memory. Unlike main memory, scratchpad memory is a part of processor unit. Fig below shows processor unit with scratchpad memory.

As shown in the fig below, Scratchpad memory has eight registers, R_0 through R_7 of 4 bits each. Therefore, the scratchpad memory can be specified as 8×4 RAM. —LM—

Registers R_0 through R_7 are assigned addresses 0 through 7. Since there are eight addresses, an address can be specified with three bits. For ex: address for

$R_0 = M[000]$ address for $R_3 = M[011]$, $R_7 = M[111]$ etc

So to perform operation $R_3 \leftarrow R_1 + R_2$ Control unit has to provide signals & select i/p's for these micro operations in sequence. Control function T_1 has to supply an address 001 to the memory & to activate the read & load A inputs.

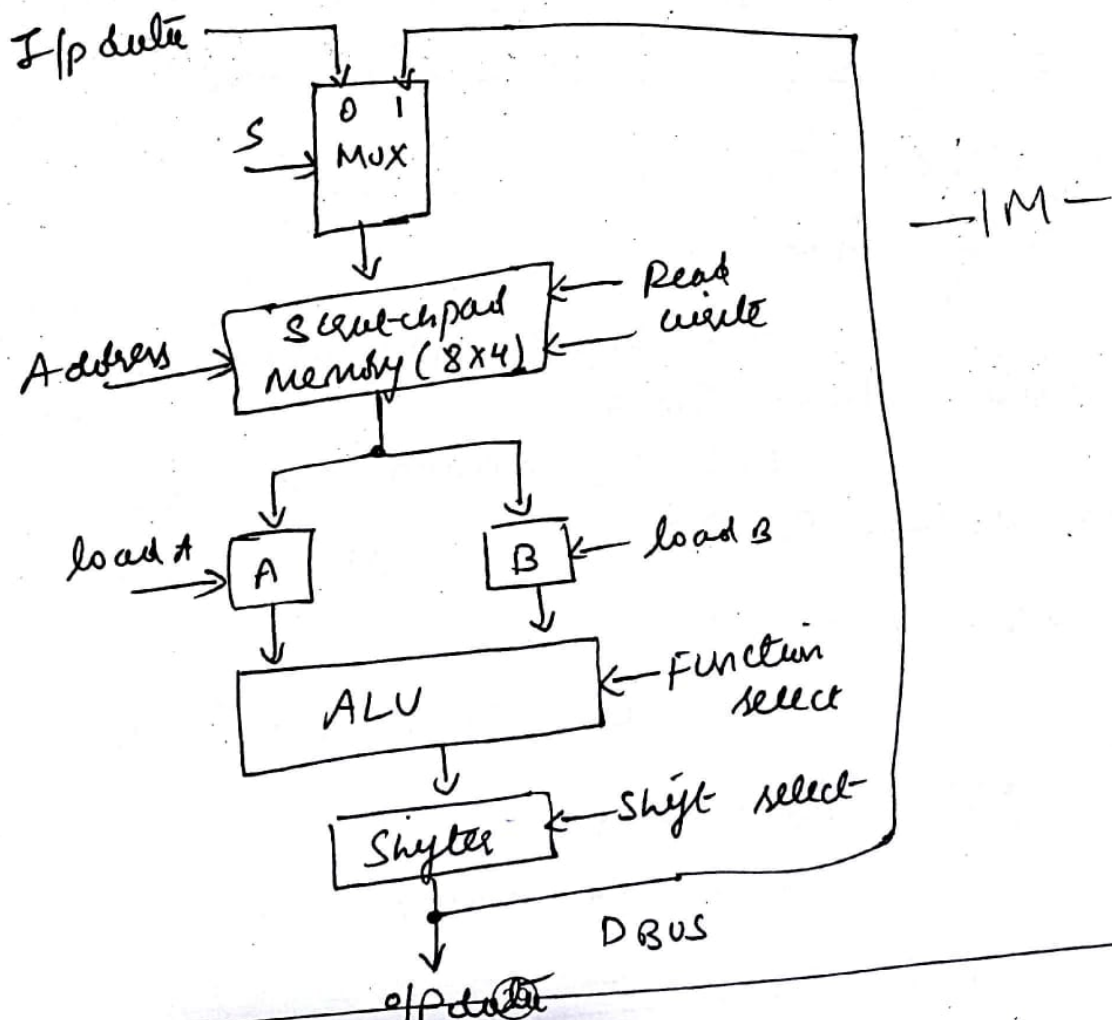
Control function T_2 has to supply an address 010 to the memory & to activate the read & load B inputs.

Control function T_3 has to supply the function select input to the ALU & Shift select i/p to Shifter to perform add operation without Shift. The control function T_3 also has to supply an address 011 to the memory & select input to the Multiplexer to select the o/p of the Shifter for the MUX, & to activate the memory write i/p.

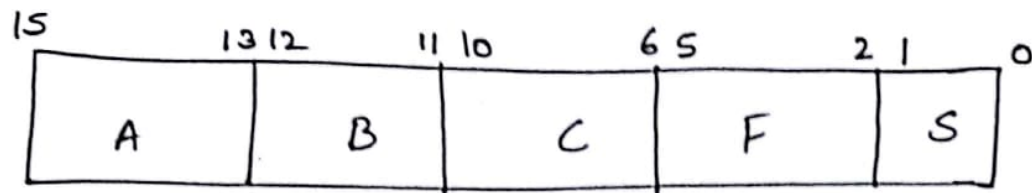
$T_1: A \leftarrow M[001]$

$T_2: B \leftarrow M[010]$

$T_3: M[011] \leftarrow A+B$



5)



A: $A_2 A_1 A_0$ — Address of specific register in the memory unit

B: B_0 — Read
 B_1 — write — 2M —

C: $C_1 C_0$ — MS_1, MS_0 — Select the Source for A
 Register 00 — Status reg
 01 — Shifter o/p
 10 — I/p data
 11 — Memory reg

C_2 : Load A

C_3 : Load B

C_4 — Sin — Serial data i/p for Shifter

F — $F_3 F_2 F_1 F_0$ — Function select inputs for ALU

S — $SL_1 SL_0$ — Function select inputs for Shifter

1) $R_1 \leftarrow Shl R_2$ — 2M —

Step 1: $A \leftarrow R_2$

010 01 00111 0000 01

Step 2: $R_1 \leftarrow Shl A$

001 10 00000 0000 00

2) $R_2 \leftarrow Shl R_6$ — 2M —

Step 1: $A \leftarrow R_6$

110 01 00111 0000 01

Step 2: $R_2 \leftarrow Shl A$

010 10 00000 0000 11

$$iv) R_1 \leftarrow R_5 - R_7 \quad - 2M -$$

$$\begin{array}{l} \text{Step 1: } A \leftarrow R_5 \quad 101 \quad 01 \quad 00111 \quad 0000 \quad 01 \\ \text{Step 2: } B \leftarrow R_7 \quad 111 \quad 01 \quad 01011 \quad 0000 \quad 01 \\ \text{Step 3: } R_1 \leftarrow A - B \quad 001 \quad 10 \quad 00000 \quad 0101 \quad 01 \end{array}$$

$$v) R_3 \leftarrow R_5 \times R_5 \quad - 2M -$$

$$\begin{array}{l} \text{Step 1: } A \leftarrow R_5 \quad 101 \quad 01 \quad 00111 \quad 0000 \quad 01 \\ \text{Step 2: } B \leftarrow R_5 \quad 101 \quad 01 \quad 01011 \quad 0000 \quad 01 \\ \text{Step 3: } R_3 \leftarrow A \oplus B \quad 011 \quad 10 \quad 00000 \quad 1010 \quad 01 \end{array}$$

$$iii) R_1 \leftarrow \text{CRC } R_2 \quad (\text{Circular right shift with carry})$$

Step 1:

$$A \leftarrow R_2$$

$$010 \quad 01 \quad 00111 \quad 0000 \quad 01$$

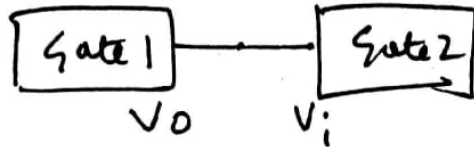
Step 2:

$$R_1 \leftarrow \text{CRC } A$$

$$001 \quad 10 \quad 00000 \quad \underline{0000} \quad 01$$

10 - CRC

If the node is at Logic HIGH, $V_{OH} > V_{IH}$



If the node is at Logic Low, $V_{OL} < V_{IL}$
(gate 1)

$$(NM)_H = V_{OH} - V_{IH} = 4 - 3 = 1V$$

$$(NM)_L = V_{IL} - V_{OL} = 1.5 - 1 = 0.5V$$

Actual Noise margin is minimum of both

$$NM = \min(NM_H, NM_L)$$

$$NM = 0.5V$$

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R. V. COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
III Semester B. E. Examinations Nov/Dec-16
Electronics and Communication Engineering
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ logic IC family has the best speed-power product.	01
	1.2	Which are the adjacent cell numbers of cell "8" in a four-variable K-map?	01
	1.3	Find the minimum SOP for the function $f(a,b,c) = M_2.M_4$ using K-map.	02
	1.4	Prove that the sum of all minterms of a Boolean function of 3 variables is one.	02
	1.5	Write the Boolean function realized by the logic circuit shown in figure 1.5.	
		Figure 1.5	02
	1.6	Design a combinational circuit with three inputs ; a, b and c and one output: F . The output is '1' when the binary value of the input is less than 3, the output is '0' otherwise.	02
	1.7	Design a combinational circuit that compares two 4-bit numbers to check if they are equal. The circuit output is equal to '1' if the two numbers are equal and '0' otherwise.	02
	1.8	Construct a T flip-flop using a D flip-flop.	01
	1.9	Write the characteristic equation for the complementary output of a $J - K$ flip-flop.	02
	1.10	What will be the type of counter if the complement outputs of the flip-flops are connected to the clock for a binary ripple counter?	01
	1.11	A flip-flop has a $20 - ns$ delay from the time CP input goes from 1 to 0 to the time the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency the counter can operate at reliably?	02
	1.12	What is the true/complement, one/zero element circuit. Illustrate it.	02

PART-B

2	a	Draw a <i>NAND</i> logic diagram that implements the complement of the following function: $F(A, B, C, D) = \sum(0, 1, 2, 3, 4, 8, 9, 12)$	05
	b	Explain the operation of a totem-pole output configuration for <i>TTL</i> gate.	06
	c	A quad two-input <i>NAND</i> gate has the following parameters: $I_{OH}(\max) = 0.4 \text{ mA}$, $V_{OH}(\min) = 2.7 \text{ V}$, $V_{IH}(\min) = 2 \text{ V}$, $V_{IL}(\max) = 0.8 \text{ V}$, $V_{OL}(\max) = 0.4 \text{ V}$, $I_{OL}(\max) = 8 \text{ mA}$, $I_{IL}(\max) = 0.4 \text{ mA}$, $I_{IH}(\max) = 20 \mu\text{A}$, $I_{CCH}(\max) = 1.6 \text{ mA}$, $I_{CCL}(\max) = 4.4 \text{ mA}$, $t_{PLH} = t_{PHL} = 15 \text{ ns}$ and the supply voltage range of 5 V . Determine: i) The average power dissipation of a single <i>NAND</i> gate ii) The maximum average propagation delay of a single gate iii) The <i>HIGH</i> -state noise margin and iv) The <i>LOW</i> -state noise margin.	05
OR			
3	a	Simplify the following Boolean function, using four-variable maps: $F(A, B, C, D) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$	04
	b	Explain the operation of a 2 input <i>CMOS NAND</i> gate and <i>NOR</i> gate, depicting transistor-level realizations.	06
	c	Define the following parameters: i) Fan-in ii) Fan-out iii) Power dissipation iv) Propagation delay v) Noise margin vi) Figure of merit.	06
4	a	Design a 4-bit carry look ahead adder.	06
	b	Using a decoder and external gates, design the combinational circuit defined by the following Boolean functions: $F_1 = x'y'z' + xz$ $F_2 = xy'z' + x'y$ $F_3 = x'y'z + xy$	05
	c	Implement the following Boolean function using a multiplexer. $F(A, B, C, D) = \sum(0, 2, 5, 7, 13, 14)$.	05
OR			
5	a	Design an excess-3 (<i>ABCD</i>) to binary (<i>WXYZ</i>) decoder. Treat the unused combinations of the code as don't care conditions.	08
	b	Design an Arithmetic incremter circuit using four half-adders and briefly explain.	04
	c	Construct a 4 to 16-line decoder with enable input, using five 2 – 4 line decoders.	04
6	a	A flip-flop has four operations: clear to 0, no change, complement and set to 1, when the inputs <i>P</i> and <i>N</i> are 00, 01, 10 and 11 respectively: i) Tabulate the characteristic table ii) Derive the characteristic equation iii) Tabulate the excitation table iv) Show how the <i>PN</i> flip-flop can be converted to a <i>D</i> -flip-flop.	05

b Design a sequential circuit with two *D* flip-flops *A* and *B* and one input *x*-in. When *x*-in = 0, the state of the circuit remains same, otherwise the circuit goes through the state transitions from 00 to 01, to 11, to 10, back to 00, and repeats. 04

c A sequential circuit has two *JK* flip-flops *A* and *B*, two inputs *x* and *y* and one output *z*. The flip-flop input equations and circuit output equations are :

$$J_A = Bx + B'y'$$

$$J_B = A'x$$

$$K_A = B'xy'$$

$$K_B = A + xy'$$

$$Z = Ax'y' + Bx'y'$$

- Draw the logic diagram of the circuit
- Tabulate the state table
- Derive the state equations for *A* and *B*.

07

OR

7 a A sequential circuit with two *D* flip flops *A* and *B*; two inputs *x* and *y* and one output *Z* is specified by the following next state and output equations:

$$A(t+1) = xy' + xB$$

$$B(t+1) = xA + xB'$$

$$Z = A.$$

- Draw the logic diagram of the circuit
- List the state table for the sequential circuit
- Draw the corresponding state diagrams.

06

b Derive the state table and state diagram of the sequential circuit shown in the fig 7b. Explain function that the circuit performs.

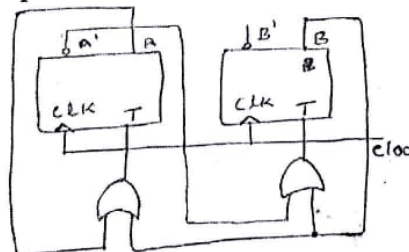


Figure 7b

04

c Design a counter with the following binary sequence: 0,4,2,1,6 and repeat, using *JK* flip-flops. 06

8 a Design a 4-bit synchronous binary up/down counter using *T* flip flops. 06

b Draw the logic diagram of a four bit register with four *D* flip-flops and 4 × 1 multiplexers with mode selection inputs *S*₁ and *S*₀. The register operates according to the function table (8b).

Table 8(b)

<i>S</i> ₁	<i>S</i> ₀	Register operation
0	0	No change
0	1	Complement the four outputs
1	0	clear register
1	1	load parallel data

06

c

Design the sequential circuit with state table(8c).

Present State		Input	Next State		Output
A_1	A_2	x	A_1	A_2	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1

OR

9 a

The serial adder of fig 9a uses two 4-bit shift registers. Register A holds the binary number 0101 and register B holds 0111. The carry flip-flop Q is initially cleared. List the binary values in register A and flip-flop Q after each shift.

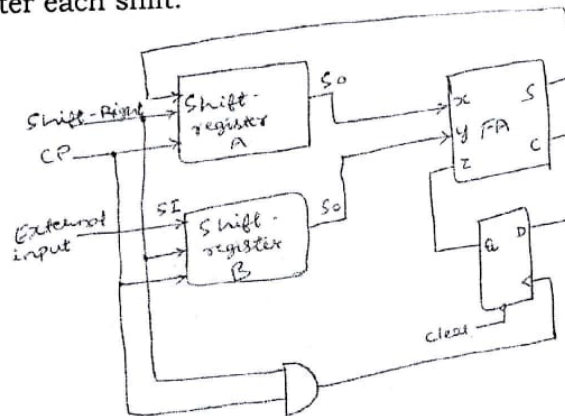


Figure 9a

b

What is the difference between Ring counter and Johnson counter.
Design a 4-bit Johnson counter.

10 a

Draw the block diagram of a processor unit that uses a scratch pad memory. Illustrate its usage to perform the operation $R_1 \leftarrow R_2 + R_3$.

b

What are status bits in a status register? How are these bits set or cleared?

c

Show how a 4 bit combinational logic shifter is constructed. Explain the operation of the shifter.

OR

11 a

Design an adder/subtractor circuit with one selection bit, S and two inputs A and B . when $S = 0$ the circuit should perform $(A + B)$ and when $S = 1$, the circuit should perform $(A - B)$ by taking 2's complement of B .

b

Draw the block diagram of a processor and describe the functions of control variables for the processor.

USN

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R. V. COLLEGE OF ENGINEERING
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 III Semester B. E. Examinations Nov/Dec-14
Electronics and Communication Engineering
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

- | | | | | | |
|---|-----|---|----|---|-----|
| 1 | 1.1 | Show the logic diagram of realizing an inverter using a 2 input NAND gate. | 01 | 2 | 1/2 |
| | 1.2 | If the output high current of a gate is 500mA and its input high current is 50mA. The fan-out is _____. | 01 | 2 | 1/2 |
| | 1.3 | What is current sourcing? | 01 | 4 | 1/2 |
| | 1.4 | The maxterm corresponding to decimal 10 is _____. | 01 | 3 | 1/2 |
| | 1.5 | Why Emitter Coupled Logic is called Current Mode Logic? | 01 | 2 | 1/2 |
| | 1.6 | What is the Boolean function realized by 8:1 MUX shown in fig 1.6. | | | |

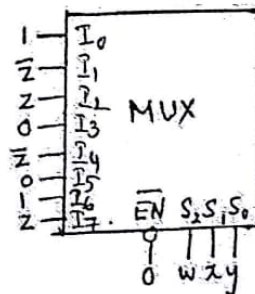


Fig 1.6

- | | | | | | |
|--|------|--|----|----|-----|
| | 1.7 | What is race around condition in a JK flipflop? | 02 | 4 | 1/2 |
| | 1.8 | What is the function of LT pin of 7447 BCD to seven segment decoder? | 01 | 4 | 1/2 |
| | 1.9 | The Q-output of SR FF is 1. The output does not change when clock pulse is applied. The inputs S and R will be respectively _____. | 01 | 13 | 1/2 |
| | 1.10 | If a counter having 10 FFs is initially at 0. The value of the count it holds after 2060 pulses is _____. | 01 | 15 | 1/3 |
| | 1.11 | A sequential circuit consists of 3FFs A, B and C and two inputs x and y. The FF C is defined by state equation as given below is implemented using JK - FF. The input functions of the FF are, _____.
$C(t+1) = C_x + AB$ | 02 | 13 | 1/4 |
| | 1.12 | A ring counter consists of n flip-flops, the number of states is equal to _____. | 01 | 12 | 1/2 |

- 1.13 The initial content of the 4-bit serial -in-parallel -out right shift register shown in figure 1.13 is 0110. After three clock pulses are applied the contents of the shift register will be _____.

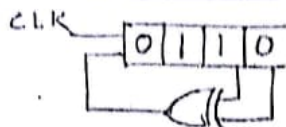


Fig 1.13

- 1.14 A ripple counter consists of 10 FFs each with a propagation delay of 10ns. What is the maximum clock frequency for reliable operation of the counter? 02
- 1.15 What is an accumulator? 01
- 1.16 Define opcode of a processor. 01
- 1.17 The address bus width and data bus of a scratchpad memory of size 512 × 8 is _____ and _____. 01

PART-B

- 2 a With a diagram, explain the operation of a CMOS NOR gate. 05
- b Find the simplified form of the following Boolean function and show its implementation: $F(A, B, C, D, E) = \sum(1, 3, 7, 8, 12, 13, 14, 15, 18, 19, 24, 25, 29)$ 07
- c What is noise margin? Calculate the noise margin of TTL and CMOS gates. Comment on the result.

Specifications:

	CMOS	TTL
$V_{OH(min)}$	4.4V	2.4V
$V_{IH(min)}$	3.5V	2V
$V_{OL(max)}$	0.33V	0.4V
$V_{IL(max)}$	1.5V	0.8V

OR

- a Describe the operation of ECL OR gate with a diagram. 04
- b A 74LS00 NAND gate output is driving three 74LS00 NAND gates and two 74AS00 NAND gates. Determine if there is a loading problem.

Specifications:

	74LS00	74AS00
I_{OH}	-0.4mA	-2mA
I_{IH}	20μA	20μA
I_{OL}	8mA	20mA
I_{IL}	-0.4mA	-0.5mA

- c A four bit binary number is represented as $A_3A_2A_1A_0$, where A_0 is the LSB. Design a logic circuit that will produce HIGH output whenever the binary number is greater than 0010 and less than 1100. 06

- a Show the construction of 4 bit carry look ahead adder. 08
- b Show realization of 5 × 32 decoder with truth table using 3 × 8 decoders. 08

OR

- 5 a Write the truth table for 3 bit binary to gray code conversion. Show the realization using 4:1 MUXs.
- b A magnitude comparator has $A(A_3A_2A_1A_0)$ and $B(B_3B_2B_1B_0)$ as inputs. Obtain the Boolean functions for $A > B$, $A < B$ and $A = B$. Show the realization using basic gates.

06

10

- 5 a Describe the working of a JK master slave FF with timing diagrams.
- b Design and show the realization of a sequential circuit characterized by the state diagram shown in Fig.6b. Apply state reduction and state assignment.

06

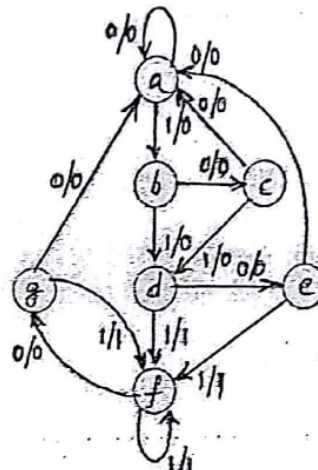


Fig. 6b

10

OR

- 7 a Realize a 3 bit ripple down counter using T flip-flops.
- b Obtain the state table, state diagram and the state equation for the sequential circuit shown in Fig. 7b:

06

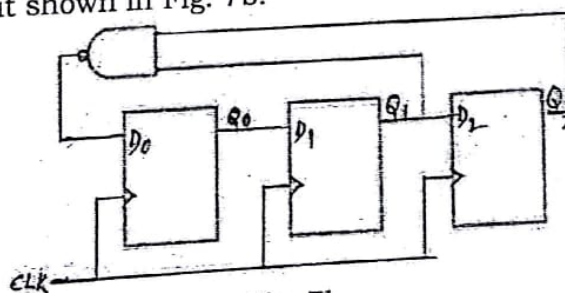


Fig. 7b

10

- 8 a Show the realization of a mod-90 counter using 7490 decade counter.
- b Design a 4 bit bidirectional shift register with parallel load. Write the function table showing different operations.

06

10

OR

- 9 a Show the realization of a synchronous 4-bit binary up down counter using JK FFs. Show the timing diagrams for up counting operation.
- b Show the realization of a 10 state Johnson counter along with the state diagram.

10

06

[illegible]

R. V. COLLEGE OF ENGINEERING
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Electronics and Communication Engineering
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

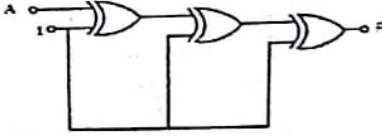
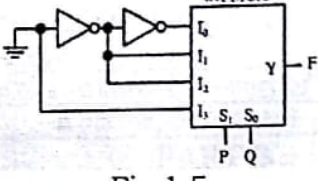
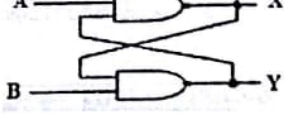
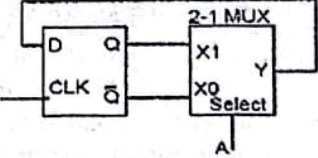
Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	$(CAFE.94)_{16} = (\quad)_8$	01
	1.2	Write the decimal number 945 in the following codes: i) 8421 code = _____ ii) 5211 code = _____.	01
	1.3	Given that $16_{10} = 100_b$, the value of $b =$ _____.	01
	1.4	For the circuit shown in fig 1.4, the output F is given by _____.	
			01
	1.5	The logic function implemented by the circuit shown in fig.1.5 is _____.	
			01
	1.6	In fig 1.6, $A = 1$ and $B = 1$. The input B is now replaced with a sequence 101010..., the outputs X and Y will be _____ and _____.	
			02
	1.7	The present output Q_n of an edge triggered JK flip-flop is logic '0'. If $j = 1$, then Q_{n+1} is _____.	01
	1.8	The output frequency of a decade counter that is clocked from a 50 KHz signal is _____.	01
	1.9	Draw the state diagram for the logic circuit shown in fig 1.9.	
			02

c	Define fan-out and fan-in. Refer to the 74ALS20 NAND gate specification given below and determine how many 74ALS20 NAND gates can be driven by the output of another 74 ALS20 NAND gates. <table border="1" data-bbox="531 324 1027 421"> <tr> <td>$I_{OH(max)}$</td> <td>$I_{IH(max)}$</td> <td>$I_{OL(max)}$</td> <td>$I_{IL(max)}$</td> </tr> <tr> <td>0.4 mA</td> <td>20 μA</td> <td>8 mA</td> <td>0.1 mA</td> </tr> </table>	$I_{OH(max)}$	$I_{IH(max)}$	$I_{OL(max)}$	$I_{IL(max)}$	0.4 mA	20 μ A	8 mA	0.1 mA	05
$I_{OH(max)}$	$I_{IH(max)}$	$I_{OL(max)}$	$I_{IL(max)}$							
0.4 mA	20 μ A	8 mA	0.1 mA							
4	a Design a 4-bit carry-look-ahead adder. b Implement the following Boolean function using an 8:1 multiplexer. $F(A, B, C, D) = A\bar{B} + BD + \bar{B}C\bar{D}$ c Design a logic circuit for 2×2 binary multiplier using AND gates and binary adders. OR	06 05 05								
5	a Design a 4-bit BCD adder that adds two BCD digits and produces a sum digit in BCD. b Construct 4- to-16 line decoder with five 2- to- 4 line decoder having enable inputs. c Write the truth table of an octal-to-binary priority encoder with LSB input (I_0) having the highest priority. Draw its block diagram having an output V to indicate that at least one of the inputs is present. What will be the value of the four outputs, if inputs I_1, I_3 and I_7 are at 1 at the same time?	06 05 05								
6	a A PN flip-flop has four operations: clear to 0, no change, complement and set to 1, when inputs P and N are 00, 01, 10 and 11 respectively. i) Tabulate the characteristic table ii) Derive the characteristic equation iii) Tabulate the excitation table iv) Show how the PN flip-flop can be converted to a D flip-flop. b A sequential circuit has one input and one output. The state diagram is shown in fig 6 b. Design a sequential circuit using D flip-flop. <pre> graph TD 001((001)) -- "0,0" --> 001 001 -- "1,1" --> 100((100)) 100 -- "1,0" --> 011((011)) 011 -- "0,0" --> 011 011 -- "1,1" --> 010((010)) 010 -- "0,0" --> 010 010 -- "1,1" --> 000((000)) 000 -- "1,1" --> 000 000 -- "0,0" --> 110((110)) 110 -- "0,0" --> 110 110 -- "1,1" --> 011 </pre> Fig 6b	05 06								
7	c A sequential circuit with 4 flip-flops (A, B, C, D) is characterized by state equation as given below: $A_{n+1} = A'BC' + A'B'C + ACD' + AC'D$ $B_{n+1} = AD' + A'BD$ $C_{n+1} = A$ $D_{n+1} = D'$ (' indicates complement of a variable) Where $A_{n+1}, B_{n+1}, C_{n+1}$ and D_{n+1} and A, B, C and D are next states and present states of FFs. Design a sequential circuit using JK Flip-Flops. OR a Show the realization of a JK master slave flip-flop using NAND gates. Explain toggling condition using input and output waveforms.	05 05								

b	A clocked sequential circuit has three states, A, B and C and one input X. As long as the input X is '0', the circuit alternates between the states A and B. If the input X becomes '1' (either in state A or in state B), the circuit goes to state C and remains in state C as long as X continues to be 1. The circuit returns to state A, if the input becomes '0' once again and from then on repeats its behavior. Assume that the state assignments are $A = 00, B = 01$ and $C = 10$. i) Draw the state diagram of the circuit. ii) Give the state table of the circuit. iii) Draw the circuit using JK flip flops.																																																						
c	Design a synchronous counter with the following Sequence: 0, 1, 3, 7, 6, 4 and repeat. Use T flip-flops.																																																						
8	a Design mod-32 ripple counter. If the propagation delay of each flip-flop is 20 nsec, determine the maximum clock frequency to be used. b Design a 4-bit shift register with a parallel load using D flip-flops. There are two control inputs: shift and load. When shift = 1, the contents of the register are shifted by one position right. New data is transferred into the register when load = 1 and shift = 0. If both the control inputs are equal to 0, the contents of the register do not change. Write the functional table showing the different operations. c Design a sequence generator for the sequence (0111) using JK flip-flops. OR																																																						
9	a Design a 4-bit BCD synchronous counter using JK flip flops and draw its timing diagrams. b Design a serial adder using a sequential logic procedure (JK FFs). Draw the state table and the state diagram for the same. c Design a 6-bit Johnson counter and draw its state diagram.																																																						
10	a Describe with a suitable diagram the bus organization of a processor register with ALU. b Design an arithmetic circuit with one selection variable S and two data inputs A and B. When $S = 1$, circuit should perform the addition operation $F = A + B$. When $S = 0$, the circuit should perform the increment operation $F = A + 1$. Draw the logic diagram of a typical stage. c Write a descriptive note on status register. OR																																																						
11	a Design an arithmetic circuit with two selection variables s_1 and s_0, which generates the arithmetic operation shown in table 1. Modify the arithmetic circuit into an ALU with mode select variable S_2. When $S_2 = 0$, the ALU is identical to the arithmetic circuit, when $S_2 = 1$, it generates the logic functions according to table 2. Draw the logic diagram of two typical stages. <table border="1" data-bbox="378 1760 1378 1986"> <thead> <tr> <th colspan="5">Table 1: Arithmetic operations</th> <th colspan="4">Table 2: Logical operations</th> </tr> <tr> <th>S_2</th> <th>S_1</th> <th>S_0</th> <th>$C_{in} = 0$</th> <th>$C_{in} = 1$</th> <th>S_2</th> <th>S_1</th> <th>S_0</th> <th>$C_{in} = 0$</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0</td> <td>0</td> <td>$F = A$</td> <td>$F = A + 1$</td> <td>1</td> <td>0</td> <td>0</td> <td>$F = A \vee B$</td> </tr> <tr> <td>0</td> <td>0</td> <td>1</td> <td>$F = A + B$</td> <td>$F = A + B + 1$</td> <td>1</td> <td>0</td> <td>1</td> <td>$F = A \oplus B$</td> </tr> <tr> <td>0</td> <td>1</td> <td>0</td> <td>$F = A - B - 1$</td> <td>$F = A - B$</td> <td>1</td> <td>1</td> <td>0</td> <td>$F = A \wedge B$</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> <td>$F = A - 1$</td> <td>$F = A$</td> <td>1</td> <td>1</td> <td>1</td> <td>$F = A$</td> </tr> </tbody> </table> b Draw the logic diagram of a 4-bit combinational logic shifter.	Table 1: Arithmetic operations					Table 2: Logical operations				S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	S_2	S_1	S_0	$C_{in} = 0$	0	0	0	$F = A$	$F = A + 1$	1	0	0	$F = A \vee B$	0	0	1	$F = A + B$	$F = A + B + 1$	1	0	1	$F = A \oplus B$	0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	$F = A \wedge B$	0	1	1	$F = A - 1$	$F = A$	1	1	1	$F = A$
Table 1: Arithmetic operations					Table 2: Logical operations																																																		
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0	0	1	$F = A + B$	$F = A + B + 1$	1	0	1	$F = A \oplus B$																																															
0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	$F = A \wedge B$																																															
0	1	1	$F = A - 1$	$F = A$	1	1	1	$F = A$																																															

USN

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Dec-13/Jan-14

Electronics and Communication Engineering

ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1

1.1

Arrange CMOS, ECL and TTL families in terms of increasing propagation delay.

01

1.2

A certain digital system design uses negative logic; what do the following function tables then represent?

a)

A	B	y
L	L	H
L	H	H
H	L	H
H	H	L

b)

A	B	y
L	L	L
L	H	H
H	L	H
H	H	H

01

1.3

Find the figure-of-merit of the following gates and choose the best gate to use.

Gate	Propagation delay	Power dissipation	Figure of merit
Standard TTL(G_1)	10ns	10mW	
Schottky TTL (G_2)	03ns	19mW	
High speed TTL(G_3)	06ns	22mW	

02

1.4

In a 5-variable K-map cell-3 is adjacent to _____, _____, _____, _____ and cell-30 to _____, _____, _____, _____.

01

1.5

A 4-bit parallel adder uses all TTL gates whose gate delays range from 8ns to 10ns. The 2nd stage produces sum S_2 after _____ delay and carry C_2 after _____ delay.

01

1.6

The POS expression for the SOP expression is _____.

$F_{(ABCD)} = \bar{B}\bar{D} + \bar{B}\bar{C} + \bar{A}\bar{C}D$

02

1.7

The minimum number of universal gates required to implement the POS expression, $\bar{y} = (A + \bar{B})(\bar{A} + B)$.

01

1.8

The minimum number of 4×1 multiplexers required to implement a 16×1 multiplexer is _____.

01

1.9

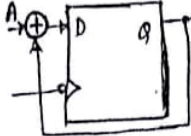
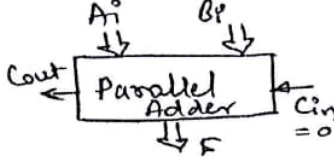
A 2×1 multiplexer uses TTL basic gates with a worst gate delay of 11ns; the overall propagation delay of the multiplexer is _____.

02

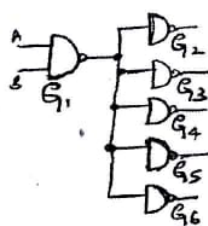
1.10

The number of 3×8 decoders needed to realize a 4×16 decoder is _____.

01

1.11	The time in which input logic level must be held constant after the application of clock is _____.	01
1.12	The analog equivalent of a latch is _____.	01
1.13	What function is performed by the circuit of fig 1.13.	01
1.14	Write the function table for the circuit of fig 1.13.	
	 Fig 1.13	01
1.15	What is the function performed by the block diagram in fig 1.15 with all B_i are logical 1's?	01
1.16	If $C_{in} = 1$ in fig 1.15, then what is the function performed?	
	 Fig 1.15	01
1.17	A 2:1 MUX can be designed using _____ half adders.	01

PART-B

2	a	Construct a 4×1 multiplexer using transition gates and show how one of the four inputs is routed on to the output.	08															
	b	Simplify using K -maps and implement $f(ABCDE) = \sum m(0,1,4,5,6,7,9,12,13,14,15,16,17,20,21,22,23,25,29)$.	08															
		OR																
3	a	Simplify the following Boolean function using K -map and obtain the expression. $f(abcde) = \sum m(8,12,13,18,19,21,22,24,25,28,30,31) + \sum \phi(1,2,4,6,7,11,26)$	07															
	b	Construct a two input $CMOS$ NOR gate and explain its working.	05															
	c	In the arrangement shown in fig 3c determine if there is any loading problem. The current specifications are as given below:																
		<table border="1"> <thead> <tr> <th></th><th>I_{OH}</th><th>I_{IH}</th><th>I_{OL}</th><th>I_{IL}</th></tr> </thead> <tbody> <tr> <td>74LS00</td><td>$-0.4mA$</td><td>$20\mu A$</td><td>$8mA$</td><td>$-0.4mA$</td></tr> <tr> <td>74AS00</td><td>$-2.0mA$</td><td>$20\mu A$</td><td>$20mA$</td><td>$-0.5mA$</td></tr> </tbody> </table>		I_{OH}	I_{IH}	I_{OL}	I_{IL}	74LS00	$-0.4mA$	$20\mu A$	$8mA$	$-0.4mA$	74AS00	$-2.0mA$	$20\mu A$	$20mA$	$-0.5mA$	
	I_{OH}	I_{IH}	I_{OL}	I_{IL}														
74LS00	$-0.4mA$	$20\mu A$	$8mA$	$-0.4mA$														
74AS00	$-2.0mA$	$20\mu A$	$20mA$	$-0.5mA$														
		 Fig 3c G_1 to $G_4 \rightarrow 74LS00$ $G_5 - G_6 \rightarrow 74AS00$	04															

4	a Design a combinational circuit that fits into the black box in fig 4a. The circuit has to product the binary output 'two greater' than the input, when input is 0,1,2,3. When the binary input is 4,5,6,7, the output is two less than the input. Fig 4a	06
	b Design and implement a 2-bit binary multiplier.	06
	c What is a multiplexer? With a neat logic diagram, explain the working of a 4 × 1 multiplexer.	04
	OR	
5	a Design a 4-bit 2's complement subtractor using IC – 7483. The circuit should produce the final correct result for all possible inputs.	08
	b Implement $f(x,y,z) = \sum m(1,2,6,7)$ using 4 × 1 multiplexer; indicate the procedure.	04
	c What is the difference between a decoder and a multiplexer? With a neat logic diagram, explain the working of a 2 × 4 decoder.	04
6	a What is race around condition? Draw neat waveforms to explain. Suggest possible methods to eliminate this problem.	04
	b Construct mod-7 counter using IC – 7490 without using external (extra) gates; explain its working.	06
	c For the state diagram shown in fig 6c, obtain the sequential circuit.	
	Fig 6c OR	06
7	a For the state diagram shown in fig 7a, obtain the reduced state diagram. For the reduced state diagram, obtain the state table.	
	Fig 7a	09

	b	Design a 4 bit counter that counts all even sequences first and then odd sequences.	0
8	a	Design a 3 bit up-down counter using <i>JK</i> Flip Flops and explain its working.	08
	b	With neat waveforms, explain the working of a ring counter (6-bit).	08
		OR	
9	a	With a neat diagram, explain how a serial adder works. Compare a serial adder with a parallel adder.	10
	b	With neat timing wave forms, explain a 6-bit Moebi counter.	06
10	a	With the help of relevant blocks, explain how a parallel adder may be used as the basic building block in the construction of an <i>ALU</i> .	10
	b	Design a status register for an <i>ALU</i> .	06
		OR	
11	a	Using <i>JK</i> flip flops, design one typical register stage that performs the micro operations given below. $P_{11}: A \leftarrow A \vee B$ $P_{12}: \leftarrow A \wedge B$	08
	b	With a neat diagram, explain a combinational shifter.	08

SCHEME AND SOLUTION

SUBJECT CODE: ¹²EC34

SUBJECT: ADDC

Question No	Part - A :-	Marks																																													
1.1	ECL, TTL and CMOS	01																																													
1.2	(a) 2:1p NOR gate (b) 2:1p AND gate.	01																																													
1.3	Fom $\Rightarrow$ Power-delay-Product (PDP) $G_1 = 100 \text{ pJ}$ $G_2 = 57 \text{ pJ}$ and $G_3 = 132 \text{ pJ}$ $\Rightarrow G_2$ is the best gate to use	02																																													
1.4	3: <u>1</u> , <u>2</u> , <u>7</u> , <u>11</u> , <u>19</u> and 30: <u>14</u> , <u>22</u> , <u>26</u> , <u>28</u> , <u>31</u> .	01																																													
1.5	$S_2: 40 \text{ ns}$ and $C_2: 40 \text{ ns}$	01																																													
1.6	$F_{ABCD} = (\bar{A} + \bar{B})(\bar{C} + \bar{D})(\bar{B} + D)$	02																																													
1.7	$\bar{Y} = (A + \bar{B})(\bar{A} + B) \Rightarrow Y = (A\bar{B} + A\bar{B}) \Rightarrow 4 \text{ NAND gates}$	01																																													
1.8	Fire (05)	01																																													
1.9	3 stages $\Rightarrow (3 \times 11) \text{ ns} = 33 \text{ ns}$	02																																													
1.10	02 (Two)	01																																													
1.11	<table border="1"> <thead> <tr> <th>present state A</th><th>Inputs X</th><th>Y</th><th>Next state A+</th><th>Hold time</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr> <tr><td>0</td><td>0</td><td>1</td><td>1</td><td></td></tr> <tr><td>0</td><td>1</td><td>0</td><td>1</td><td></td></tr> <tr><td>0</td><td>1</td><td>1</td><td>0</td><td></td></tr> <tr><td>1</td><td>0</td><td>0</td><td>1</td><td></td></tr> <tr><td>1</td><td>0</td><td>1</td><td>0</td><td></td></tr> <tr><td>1</td><td>1</td><td>0</td><td>0</td><td></td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td></td></tr> </tbody> </table>	present state A	Inputs X	Y	Next state A+	Hold time	0	0	0	0		0	0	1	1		0	1	0	1		0	1	1	0		1	0	0	1		1	0	1	0		1	1	0	0		1	1	1	1		02
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0	0	0	0																																												
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1.12	Bistable multivibrator	01																																													
1.13	T-Flip Flop	01																																													
1.14	<table border="1"> <thead> <tr> <th>Ck</th><th>A</th><th>Q_{n+1}</th></tr> </thead> <tbody> <tr><td>0</td><td>x</td><td>Q_n</td></tr> <tr><td>↑</td><td>x</td><td>Q_n</td></tr> <tr><td>↓</td><td>0</td><td>$\bar{Q}_n$</td></tr> <tr><td>↓</td><td>1</td><td>Q_n</td></tr> </tbody> </table>	Ck	A	Q_{n+1}	0	x	Q_n	↑	x	Q_n	↓	0	$\bar{Q}_n$	↓	1	Q_n	01																														
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↓	0	$\bar{Q}_n$																																													
↓	1	Q_n																																													
1.15	What is the function performed by blocks in Fig Q1.15 with all Bi are logical 1's? $\Rightarrow$ Decrement 'A'	01																																													
1.16	Transfer 'A' $\rightarrow$ 01m																																														
1.17	02 — 01m																																														

Page No-1

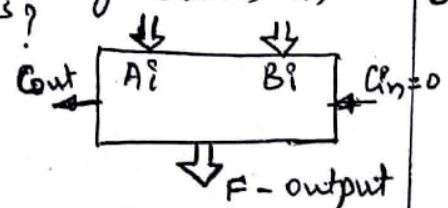


Fig Q1.15

SCHEME AND SOLUTION

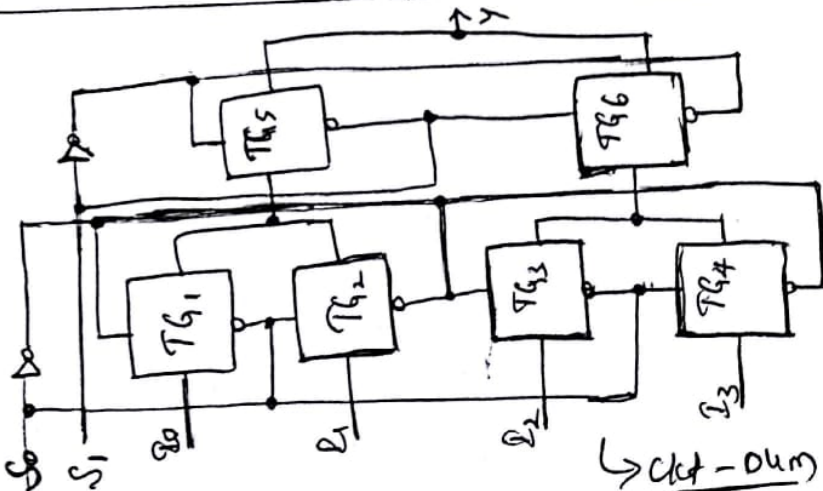
SUBJECT CODE: 12 EC 34

SUBJECT: ADDC

Question No

- PART-B :-

2 (a)



S_1, S_0	Y
0 0	Y_0
0 1	Y_1
1 0	Y_2
1 1	Y_3

→ 1m
Brief Explan → 2m

(b)

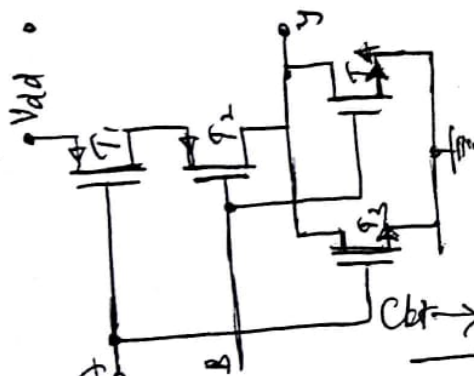
$(CABDE) = \bar{B}C + \bar{B}\bar{D} + \bar{D}E + \bar{A}C \rightarrow \text{Kmap + simplification } 04 + 02 = 06m$
Implementation → 02m

3 (a)

$f(CABDE) = b\bar{a}e + a\bar{d}e + \bar{a}b\bar{c}d + \bar{a}b\bar{c}d + \bar{a}b\bar{c}d + \bar{a}b\bar{c}d$
→ Kmap & simplification → $4 + 2 = 06m$; → Explan - 01

Function Table → 1m

(b)



A	B	Y
0	0	
0	1	
1	0	
1	1	

Brief Explan - 2m

(c)

$I_{sink} = (3 \times 10 + 2 \times 20) \mu A = 100 \mu A < I_{OL} = 0.4 \text{ mA} \rightarrow 2m$
 $I_{source} = (3 \times 0.4 + 2 \times 0.5) \text{ mA} = 2.2 \text{ mA} < I_{OL} = 8 \text{ mA} \rightarrow 2m$
⇒ No loading

4 (a)

Function table: 02m Explan: 02m Implan: 02m

(b)

Function table: 03m Implan: 03m

(c)

Mux; relationship betwn select lines & I/p lines - 01m
Mux dia: 2m and Brief Explan - 01m

5 (a)

A decoder is a de-mux, but without data I/p line; relationship betwn select & I/p lines - 01m
Dec - dia - 02m, Brief Explan - 01m

SCHEME AND SOLUTION

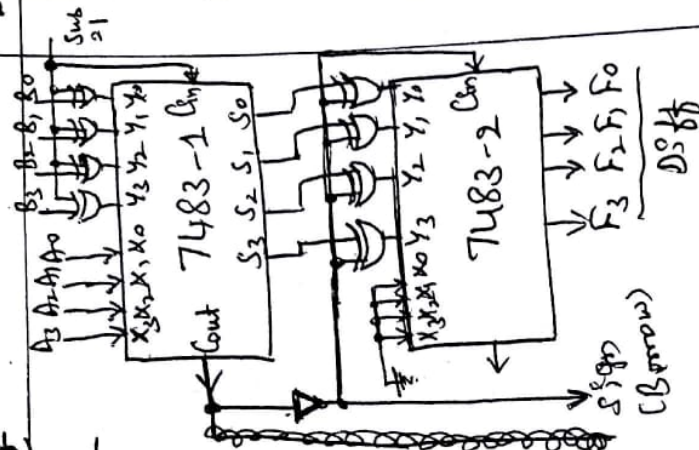
SUBJECT CODE: 12EC34

SUBJECT: ADDC

Marks

Question No

5 (a)



Ckt - 04m
Brief Explan - 03m
Example - 01m

08

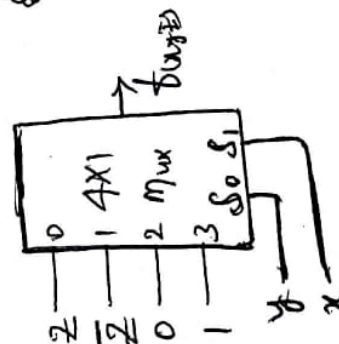
(b)

$$f(x, y, z) = \sum_m(1, 2, 6, 7)$$

Design: 02m

X \ Y \ Z	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
f=0	0	0	0	0	0	0	0	0
f=1	0	1	1	0	1	0	1	1
f=2	0	0	1	1	0	1	0	0
f=3	1	1	0	0	1	0	1	1

Imp: 02m



04

6(a) What is Race around condition? Draw neat waveforms to explain. Suggest possible methods to eliminate this problem.

(b) Construct Mod-7 counter using IC-7490 without using any extra gates; Explain its working.

(c) For the State diagram shown in

6(a) Race around + waveforms - 02m
methods - 02m

04

6(b) Ckt - 02m, TT - 02m, Explan - 02m

06

6(c) State table - 01m, Equib - 03m; Ckt - 02m

06

7(a) Reduction - 04m; Reduced dia - 01m,
State table - 02m; Brief Explan - 02m

09

(b) TT - 01m, design - 04m, brief Explan - 02m

07

SCHEME AND SOLUTION

SUBJECT CODE:

SUBJECT:

Question No	
8(a)	Dsa - 03m, TT - 02m, Working - 03m
(b)	Dsa - 02m, Waveforms - 04m - Exph. - 02m
	OR
9(a)	Dsa - 04m, Working - 04m, Comparison - 02m
(b)	Dsa - 02m, W/f - 03m - Briefing - 01m
10(a)	Logic blocks for various operations - 06m Justification with one example - 2m
(b)	Design a status register for an ALU. Diagram - 04m, Briefing - 02m
	OR
11(a)	Each - 04m 2x04 = 08
(b)	Dsa - 04m, FT - 02m, Exph. - 02m

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R. V. COLLEGE OF ENGINEERING
 Autonomous Institution affiliated to VTU
 III Semester B. E. Examinations Nov/Dec-14
Electronics and Communication Engineering
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS

Time: 03 Hours

Instructions to candidates:

Maximum Marks: 100

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Show the logic diagram of realizing an inverter using a 2 input NAND gate.	01	BT 03
	1.2	If the output high current of a gate is 500mA and its input high current is 50mA. The fan-out is _____.	01	1 1/2
	1.3	What is current sourcing?	01	2 1/2
	1.4	The maxterm corresponding to decimal 10 is _____.	01	1 1/2
	1.5	Why Emitter Coupled Logic is called Current Mode Logic?	01	3 1/2
	1.6	What is the Boolean function realized by 8:1 MUX shown in fig 1.6.	01	2 1/2
Fig 1.6				
	1.7	What is race around condition in a JK flipflop?	02	4 1/2
	1.8	What is the function of LT pin of 7447 BCD to seven segment decoder?	01	1 1/2
	1.9	The Q-output of SR FF is 1. The output does not change when clock pulse is applied. The inputs S and R will be respectively _____.	01	1 1/2
	1.10	If a counter having 10 FFs is initially at 0. The value of the count it holds after 2060 pulses is _____.	01	3 1/2
	1.11	A sequential circuit consists of 3 FFs A, B and C and two inputs x and y. The FF C is defined by state equation as given below is implemented using JK - FF. The input functions of the FF are, _____.	01	3 1/2
		$C(t+1) = C_x + AB$		
	1.12	A ring counter consists of n flip-flops, the number of states is equal to _____.	02	5 1/2
			01	2 1/2

CO BT
1,2,3 1,2

- 1.13 The initial content of the 4-bit serial -in-parallel -out right shift register shown in figure 1.13 is 0110. After three clock pulses are applied the contents of the shift register will be _____.

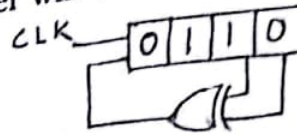


Fig 1.13

1,2,3 3

- 1.14 A ripple counter consists of 10FFs each with a propagation delay of 10ns. What is the maximum clock frequency for reliable operation of the counter?

1,2,3 1

- 1.15 What is an accumulator?

1,2,3 2

- 1.16 Define opcode of a processor.

1,2,3 3

- 1.17 The address bus width and data bus of a scratchpad memory of size 512 x 8 is _____ and _____.

PART-B

1,2,3 3

1,2,3 3

1,2,3 2

- 2 a With a diagram, explain the operation of a CMOS NOR gate.
b Find the simplified form of the following Boolean function and show its implementation: $F(A, B, C, D, E) = \sum(1, 3, 7, 8, 12, 13, 14, 15, 18, 19, 24, 25, 29)$
c What is noise margin? Calculate the noise margin of TTL and CMOS gates. Comment on the result.

Specifications:

	CMOS	TTL
$V_{OH(min)}$	4.4V	2.4V
$V_{IH(min)}$	3.5V	2V
$V_{OL(max)}$	0.33V	0.4V
$V_{IL(max)}$	1.5V	0.8V

OR

1,2,3 3

1,2,3 3

- 3 a Describe the operation of ECL OR gate with a diagram.
b A 74LS00 NAND gate output is driving three 74LS00 NAND gates and two 74AS00 NAND gates. Determine if there is a loading problem.

Specifications:

	74LS00	74AS00
I_{OH}	-0.4mA	-2mA
I_{IH}	20μA	20μA
I_{OL}	8mA	20mA
I_{IL}	-0.4mA	-0.5mA

1,2,3 6

- c A four bit binary number is represented as $A_3A_2A_1A_0$, where A_0 is the LSB. Design a logic circuit that will produce HIGH output whenever the binary number is greater than 0010 and less than 1100.

1,2,3 6

2,3 6

- 4 a Show the construction of 4 bit carry look ahead adder.
b Show realization of 5 x 32 decoder with truth table using 3 x 8 decoders.

OR

5	a	Write the truth table for 3 bit binary to gray code conversion. Show the realization using 4:1 MUXs. A magnitude comparator has $A(A_3A_2A_1A_0)$ and $B(B_3B_2B_1B_0)$ as inputs. Obtain the Boolean functions for $A > B$, $A < B$ and $A = B$. Show the realization using basic gates.	06	3, 1, 3
	b		10	3, 1, 3
6	a	Describe the working of a JK master slave FF with timing diagrams. Design and show the realization of a sequential circuit characterized by the state diagram shown in Fig.6b. Apply state reduction and state assignment.	06	2, 2, 3
	b			
		Fig. 6b	10	6, 1, 3
		OR		
7	a	Realize a 3 bit ripple down counter using T flip-flops. Obtain the state table, state diagram and the state equation for the sequential circuit shown in Fig. 7b:	06	3, 1, 2
	b			
		Fig. 7b	10	3, 1, 2
8	a	Show the realization of a mod-90 counter using 7490 decade counter. Design a 4 bit bidirectional shift register with parallel load. Write the function table showing different operations.	06	3, 1, 2
	b		10	6, 1, 2
		OR		
9	a	Show the realization of a synchronous 4- bit binary up down counter using JK FFs. Show the timing diagrams for up counting operation. Show the realization of a 10 state Johnson counter along with the state diagram.	10	3, 1, 2
	b		06	3, 1, 2

10 a

Describe the bus organization of processor registers and ALU with diagrams.

b

Design an arithmetic circuit, with two selection variables S_1 and S_0 , that generates the following arithmetic operations. Draw the logic diagram of one typical stage.

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A + B$	$F = A + B + 1$
0	1	$F = A$	$F = A + 1$
1	0	$F = \bar{B}$	$F = \bar{B} + 1$
1	1	$F = A + \bar{B}$	$F = A + \bar{B} + 1$

OR

11 a

Design an arithmetic and logic circuit, with three selection variables S_2 , S_1 and S_0 , that generates the following operations. Draw the logic diagram of one typical stage.

Arithmetic Operations:					Logical Operations:				
S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	S_2	S_1	S_0	$C_{in} = X$	
0	0	0	$F = A$	$F = A + 1$	1	0	0	$F = A \vee B$	OR
0	0	1	$F = A + B$	$F = A + B + 1$	1	0	1	$F = A \oplus B$	EX-OR
0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	$F = A \wedge B$	AND
0	1	1	$F = A - 1$	$F = A$	1	1	1	$F = \bar{A}$	Complement

b

What is a status register? Mention the importance of carry, overflow and zero flags.

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Department of Electronics and Communication Engineering

Analysis and Design of Digital circuits 12EC34

III Semester

1st Test

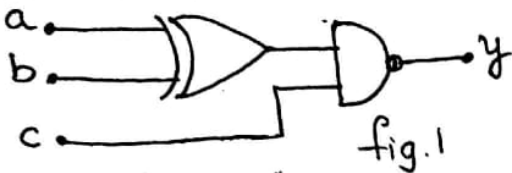
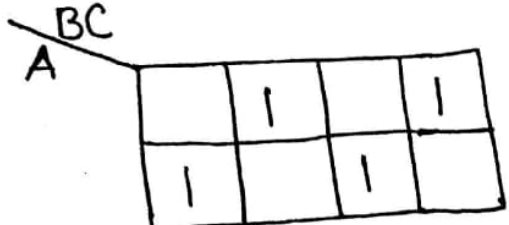
Time: 2 hours

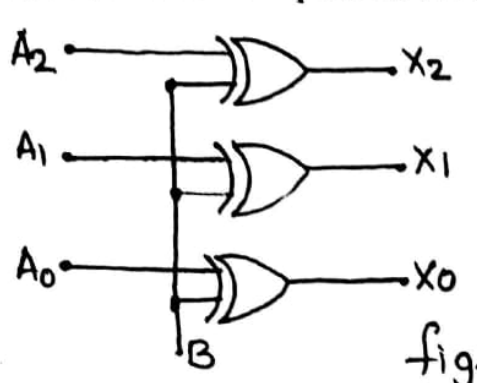
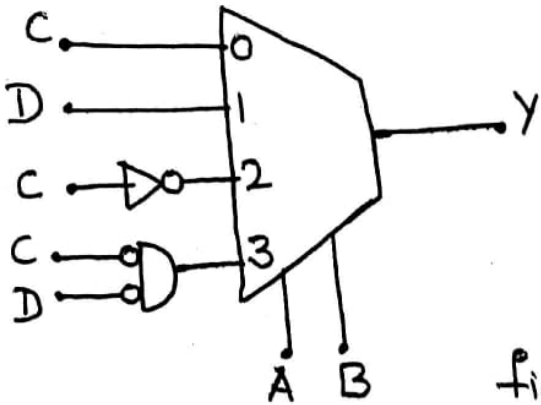
Maximum Marks: 65

Instructions to candidates:

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

PART-A

SLNo	QUESTION	M	BT	PO	CO
1	$4321.23_{(5)} = \underline{\hspace{2cm}}_{(10)}$.	1	2,3	b,c,e	1,2
2	$D00D_{(16)} = \underline{\hspace{2cm}}_{(8)}$.	1	2,3	b,c,e	1,2
3	A XNOR 1 = <u> </u> .	1	2,3,4	b,c	1
4	The minimal Boolean expression for the output of the logic circuit shown in fig1 is <u> </u>.  fig.1	1	2,3,4	b,c,e	1,2, 3,4
5	A combinational circuit has inputs A, B and C and its Karnaugh map is shown in fig2. The output of the circuit is given by <u> </u>.  fig.2	1	2,3,4	b,c,d	2,3, 4
6	For the identity $AB + A'C + BC = AB + A'C$, the dual form is <u> </u>.	1	1,2,6	b,c,k	1,2

7	The max term corresponding to decimal number 15 is _____.	1	1,2,6	b,c,k
8	In fig3, X_2, X_1, X_0 will be 1's complement of A_2, A_1, A_0, if _____.  fig3	1	1,3,4	b,j,k
9	In a certain application four inputs A, B, C, D are fed to logic circuit, producing an output which operates a relay. The relay turns on when $f(A, B, C, D) = 1$ for the following states of the input 0000, 0010, 0101, 0110, 1101 and 1110. State 1000 and 1001 do not occur, and for the remaining states the relay is off. The minimized boolean expression f is _____.	2	2,3,4	b,c,d ,f
10	A full adder can be implemented with half adders and OR gates. A four bit parallel full adder without any initial carry requires _____ half adders and _____ OR gates.	2	1,2	h,j,k
11	The boolean function realized by the logic circuit in fig.4 is _____.  fig.4	2	1,3,4	b,c,d
12	A NOR gate with two active low inputs is equivalent to _____ gate.	1	2,3,5	b,c

PART-II

Sl.No	QUESTION	M	BT	PO	CO
1	a) Convert the decimal numbers +46 and +29 into its equivalent binary numbers using 8-bit signed 2's complement representation. Then, perform the following operations: (i) (+29) + (-49) (ii) (-29) + (+49) (iii) (-29) + (-49). b) The state of a 12-bit register is 100010010111. What is its content, if it represent I. Three decimal digits in BCD. II. Three decimal digits in Excess-3 code c) If a binary number is 1001101001, then find its equivalent gray code.	6 2 1	2,3,4	a,b	1,2
2	a) Simplify $f(x,y,z) = xyz' + x'yz + xyz + x'yz'$ using Boolean laws. b) Find all the prime implicants for the following Boolean function and determine which of these are essential. $F(A,B,C,D) = \sum m(0,2,3,5,7,8,10,11,13,15)$	2 3	1,2,3	b,c, d,e	1,2 ,4, 5
3a	Simplify the following expressions to SOP and POS : (i) $F(w,x,y,z) = w'z + xz + x'y + wx'z$ (ii) $F(A,B,C,D) = AB'C + B'C'D' + BCD + ACD' + A'B'C + A'BC'D$ (iii) $F(A,B,C,D,E) = \sum m(0,1,4,5,16,17,21,25,29)$	3 3 4	1,2,3	a,b, d	1,2 ,4
4	a) Design a 1-bit full adder using minimum number of NAND gates. b) Design a SOP circuit to detect the decimal numbers 0,2,4,6 and 8 in a 4-bit 8421 code input.	5 6	4,5	a,b, c	1,2 ,3
5	a) Construct a 16X1 multiplexer with two 8X1 and one 2X1 multiplexers. b) Design a combinational circuit to multiply two 2-bit binary numbers. c) Explain how does a carry lookahead adder increases the speed with proper logical expressions.	5 4 6	1,2,3 ,4,5	a,b,	1,2 ,4

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Analysis and Design of Digital circuits 12EC34

III Semester

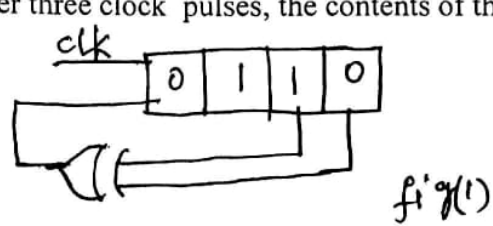
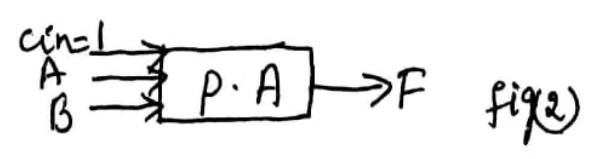
3rd Test

Time: 2 hours

Instructions to candidates:

Maximum Marks:-65

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

Sl.No	QUESTION	M	BT	PO	CO
1	The content of a 6-bit shift register is initially 110101. The content of the register is shifted left two times with serial input being 10. The content of the register after each shift is _____ and _____.	1	2,3	b,c,e	1,2
2	The initial contents of the 4 bit serial in parallel out right shift register in fig1 is 0110. After three clock pulses, the contents of the shift register is _____. 	2	2,3	b,c,e	1,2
3	An 8-bit switch -tail ring counter will go through a sequence of _____ states.	1	2,3,4	b,c	1
4	A serial adder uses two 4-bit shift registers. Register A holds a binary number 0110 and register B holds 1000. The carry flip-flop Q is initially set. List the binary values in register A and the output of flip-flop Q after each shift. Assume Sum is shifted in register A.	2	2,3,4	b,c,e	1,2, 3,4
5	The address bus width of a scratchpad memory of size 1024X8 is _____.	1	2,3,4	b,c,d	2,3, 4
6	A bus organized processor consists of 13 registers. The number of selection lines in each multiplexer and in the destination decoder are _____ and _____.	1	1,2,6	b,c,k	1,2
7	In the parallel adder of fig.2, if $C_{in}=1$ then the function performed is _____. 	1	1,2,6	b,c,k	1,2

8	In status register, Z is ____ if the contents of ALU are ____.	1	1,3,4	b,j,k	1,2
9	In an 8-bit ALU, bit V is set, if _____ is equal to 1.	1	2,3,4	b,c,d	1,2
10	In a processor with an accumulator register list the sequence of micro-operations for the operation $A \leftarrow R_1 + R_2$.	2	1,2	h,j,k	1,2
11	The input to a 4-bit parallel adder are A, B and C_{in} . If $C_{in} = 1$ and B=all 1's, then $F = \underline{\hspace{2cm}}$ and $C_{out} = \underline{\hspace{2cm}}$.	2	1,3,4	b,c,d	1,2

PART-B

PART-B

Sl.No	QUESTION	M	BT	PO	CO																																													
1	a) Design a 4-bit shift register with a parallel load using D flip-flops. There are two control inputs: shift and load. When shift = 1, the contents of the register are shifted by one position right. New data is transferred into the register when load = 1 and shift=0. If both control inputs are equal to 0 the contents of the register do not change. b) Show the realization of a ring counter with 6 states.	6 4	2,3 4	a,b	1,2																																													
2	a) Design a serial subtractor using a sequential logic procedure. (JK FFs). Draw the state table and the state diagram for the same.	10	1,2 3	b,c, d,e	1, 4, 5																																													
3	a) Design an arithmetic circuit with one selection variable S and two data inputs A and B. When S=0, circuit should performs the addition operation $F=A+B$. When S=1, the circuit should performs the increment operation $F=A+1$. b) Draw the logic diagram of a 4-bit combinational logic shifter.	6 4	1,2 3	a,b, d	1																																													
4	Design an arithmetic and logic circuit with three selection variables s_2, s_1, and s_0, that generates the following operations. Draw the logic diagram of one typical stage. Arithmetic operations: <table border="1"> <thead> <tr> <th>s_2</th> <th>s_1</th> <th>s_0</th> <th>$C_{in}=0$</th> <th>$C_{in}=1$</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0</td> <td>0</td> <td>$F=A$</td> <td>$F=A+1$</td> </tr> <tr> <td>0</td> <td>0</td> <td>1</td> <td>$F=A+B$</td> <td>$F=A+B+1$</td> </tr> <tr> <td>0</td> <td>1</td> <td>0</td> <td>$F=A-B-1$</td> <td>$F=A-B$</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> <td>$F=A-1$</td> <td>$F=A$</td> </tr> </tbody> </table> Logical operations: <table border="1"> <thead> <tr> <th>s_2</th> <th>s_1</th> <th>s_0</th> <th>$C_{in}=0$</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>0</td> <td>0</td> <td>$F=A \vee B$</td> </tr> <tr> <td>1</td> <td>0</td> <td>1</td> <td>$F=A \oplus B$</td> </tr> <tr> <td>1</td> <td>1</td> <td>0</td> <td>$F=A \wedge B$</td> </tr> <tr> <td>1</td> <td>1</td> <td>1</td> <td>$F=A$</td> </tr> </tbody> </table>	s_2	s_1	s_0	$C_{in}=0$	$C_{in}=1$	0	0	0	$F=A$	$F=A+1$	0	0	1	$F=A+B$	$F=A+B+1$	0	1	0	$F=A-B-1$	$F=A-B$	0	1	1	$F=A-1$	$F=A$	s_2	s_1	s_0	$C_{in}=0$	1	0	0	$F=A \vee B$	1	0	1	$F=A \oplus B$	1	1	0	$F=A \wedge B$	1	1	1	$F=A$	10	4,5	a,b, c	
s_2	s_1	s_0	$C_{in}=0$	$C_{in}=1$																																														
0	0	0	$F=A$	$F=A+1$																																														
0	0	1	$F=A+B$	$F=A+B+1$																																														
0	1	0	$F=A-B-1$	$F=A-B$																																														
0	1	1	$F=A-1$	$F=A$																																														
s_2	s_1	s_0	$C_{in}=0$																																															
1	0	0	$F=A \vee B$																																															
1	0	1	$F=A \oplus B$																																															
1	1	0	$F=A \wedge B$																																															
1	1	1	$F=A$																																															
5	a) Describe with a suitable diagram the bus organization of a processor register with ALU. b) Write a note on status register.	6 4	1,2 4, 5	a,b																																														

R.V.COLLEGE OF ENGINEERING, Bangalore
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Department of Electronics and Communication Engineering
ANALYSIS & DESIGN OF DIGITAL CIRCUITS-12EC34
III Semester
2nd Test

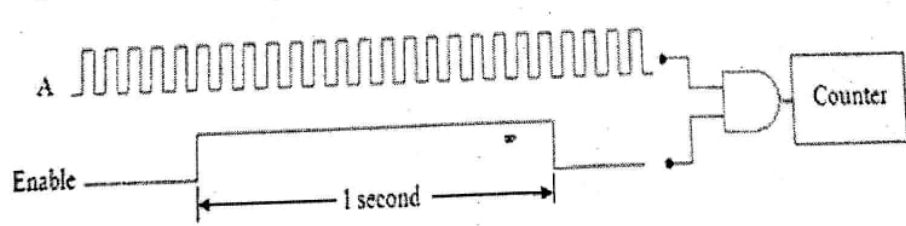
Time: 2 hours

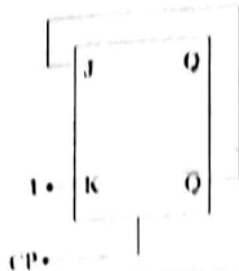
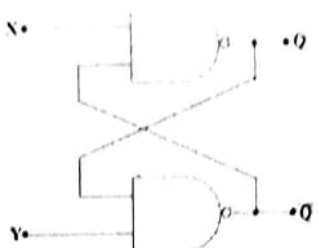
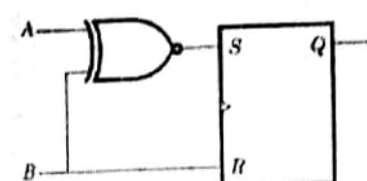
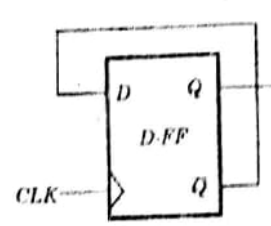
Maximum Marks: 65

Instructions to candidates:

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

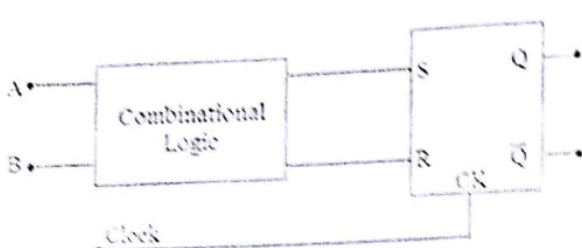
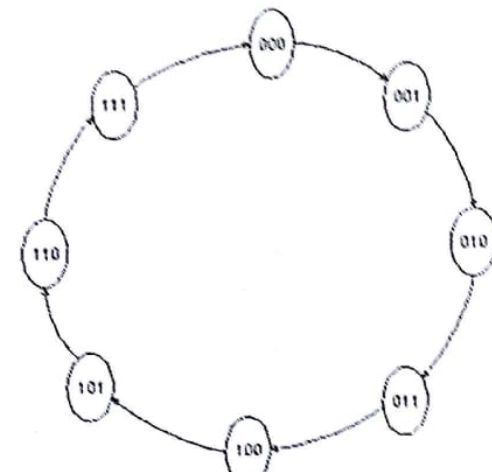
PART-A

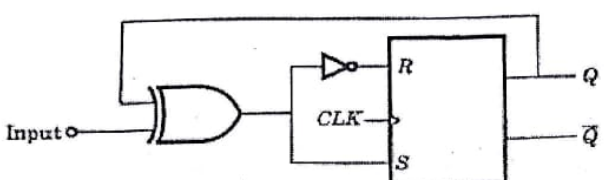
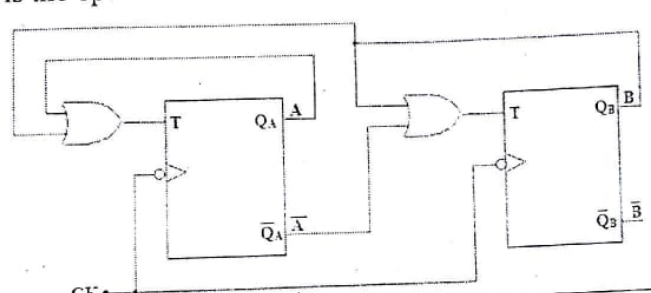
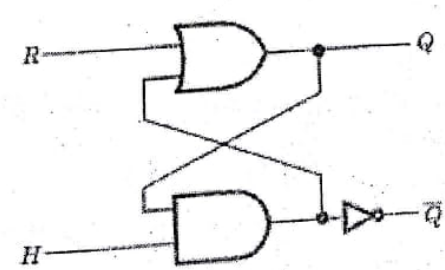
Sl.No	QUESTION	M	BT	PO	CO
1.	The Q-output of SR FF is 0. The output does not change when clock pulse is applied. The inputs S and R will be respectively _____.	1M	2,3,4,6	a,b,d ,h, k,l	2,3, 4,6
2.	The number of FFs required to build a binary counter circuit to count from 0 to 1023 is _____.	1M	1,2,3 ,4,6	a,b,d ,h, k	2,3, 4,6
3.	The correct statements in the following are, i. A FF used to store 1-bit information. ii. Race around condition occurs in JK-FF when both the inputs are 1. iii. Master-Slave configuration is used in FFs to store 2-bit information. iv. A transparent latch consists of D-type FF.	1M	1,2,3 ,4,5	a,b,d ,h,k	2,3, 4,6
4.	The diagram below shows an arrangement to measure the unknown frequency of the signal A. Where, AND gate controls the passage of signal to the digital counter. Assume the counter is initially cleared and positive edge triggered. The possible number of FFs in the counter is, 	1M	1,2,3,5	a,b,d ,h,k	2,3, 4,6
5.	A JK-FF is connected as shown below. Assuming the FF was initially cleared and then clocked for 6 pulses. The sequence at the Q output will be,	1M	1,2,3	a,b,d ,h,k	2,3, 4,6

					
6.	A sequential circuit consists of 3 FF's A, B and C and two inputs x and y. The FF A is defined by state equation as given below and is implemented using T-FF. The input functions of the FF are, $A(t+1) = xAB + y\bar{A}C + xy$	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6
7.	For FF formed from NAND gates as shown below, what is the value of X & Y for indeterminate state? 	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6
8.	The process of storing a logic-0 in a positive logic flip flop circuit is called as _____.	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6
9.	AB FF is constructed from SR FF as shown below. The expression for next state is, 	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6
10.	For the circuit shown below, what is the frequency of the output Q (in terms of clock frequency)? FF is cleared initially. 	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6
11.	Consider a circuit shown in figure. The circuit functions as,	IM	1,2,3 4,5	a,b,d h,k	2,3 4,6

PART-B

PART-B

Sl.No	QUESTION	M	BT	PO	CO															
1.	a. Show the realization of JK FF using NAND gates. Explain toggling condition using input & output waveforms. b. A sequential circuit is shown below. Determine the combinational logic for S & R. Truth Table <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Q</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>$\bar{Q}_n$</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>Q_n</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> 	A	B	Q	0	0	$\bar{Q}_n$	0	1	1	1	0	Q_n	1	1	0	5 M 5 M	1,2,3 4,5,6	a,b, d,h, k	2,3, 4,6
A	B	Q																		
0	0	$\bar{Q}_n$																		
0	1	1																		
1	0	Q_n																		
1	1	0																		
2.	a. Design mod-8 synchronous counter shown with state transitions as shown in diagram below using JK FFs. 	10 M	1,2,3 4,6	a,b, d,h, k	2,3, 4,6															
3.	Design mod-16 asynchronous down counter using JK FFs. Show the effect of propagation delay of FFs with timing diagrams. Determine rate of clock if propagation delay through each FF is 10 ns.	10M	1,2,3 4,5,6	a,b, d,h, k	2,3, 4,6															
4.	a. What is asynchronous input? Realize JK FF with PRESET & CLEAR inputs. Show input & output waveforms. b. A sequential circuit with 4 FFs (A, B, C, D) is characterized by state equations as given below. $A_{n+1} = A'BC' + A'B'C + ACD' + AC'D$, $B_{n+1} = A'C + AD' + A'BD'$, $C_{n+1} = A$, $D_{n+1} = D'$ (' indicate compliment) Where, A_{n+1}, B_{n+1}, C_{n+1}, D_{n+1} and A, B, C, D are next states & present states of FFs. Design the sequential circuit using JK FF.	4 M 6 M	1,2,3 4,6	a,b, d,h, k	2,3, 4,6															
5.	a. Describe the working of JK master slave FF with timing diagrams. b. Show the realization of SR FF using T FF.	6 M 4 M	1,2,3, 4,6	a,b, d,h, k	1,2, 3,4															

					
12.	The sequential circuit shown below consists of two T-FFs with initial state 00. What is the operation of the circuit? 	2M	1,2,3 ,4,5	a,b,d ,h,k	2,3, 4,6
13.	What is lock out condition?	1M	1,2,3 ,4,5	a,b,d ,h,k	2,3, 4,6
14.	Consider a latch circuit shown in figure below, which of the set of input is invalid for circuit, 	1M	1,2,3 ,4,5	a,b,d ,h,k	2,3, 4,6

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Department of Electronics and Communication Engineering

Analysis and Design of Digital circuits 12EC34

III Semester

3rd Test

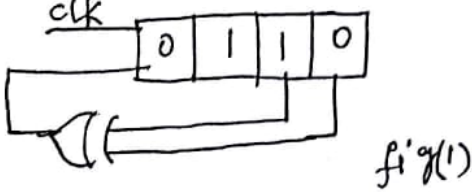
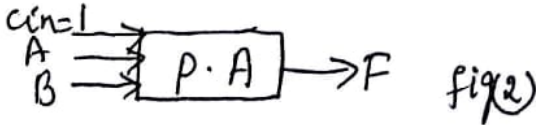
Time: 2 hours

Maximum Marks:-65

Instructions to candidates:

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

PART-A

Sl.No	QUESTION	M	BT	PO	CO
1	The content of a 6-bit shift register is initially 110101. The content of the register is shifted left two times with serial input being 10. The content of the register after each shift is _____ and _____.	1	2,3	b,c,e	1,2
2	The initial contents of the 4 bit serial in parallel out right shift register in fig1 is 0110. After three clock pulses, the contents of the shift register is _____. 	2	2,3	b,c,e	1,2
3	An 8-bit switch -tail ring counter will go through a sequence of _____ states.	1	2,3,4	b,c	1
4	A serial adder uses two 4-bit shift registers. Register A holds a binary number 0110 and register B holds 1000. The carry flip-flop Q is initially set. List the binary values in register A and the output of flip-flop Q after each shift. Assume Sum is shifted in register A.	2	2,3,4	b,c,e	1,2, 3,4
5	The address bus width of a scratchpad memory of size 1024X8 is _____.	1	2,3,4	b,c,d	2,3, 4
6	A bus organized processor consists of 13 registers. The number of selection lines in each multiplexer and in the destination decoder are _____ and _____.	1	1,2,6	b,c,k	1,2
7	In the parallel adder of fig.2, if $C_{in}=1$ then the function performed is _____. 	1	1,2,6	b,c,k	1,2

8	In status register, Z is ____ if the contents of ALU are ____.	1	1,3,4	b,j,k	1,2
9	In an 8-bit ALU, bit V is set, if ____ is equal to 1.	1	2,3,4	b,c,d	1,2
10	In a processor with an accumulator register list the sequence of micro-operations for the operation $A \leftarrow R_1 + R_2$.	2	1,2	h,j,k	1,2
11	The input to a 4-bit parallel adder are A, B and C_{in} . If $C_{in} = 1$ and B=all 1's, then $F = \underline{\hspace{2cm}}$ and $C_{out} = \underline{\hspace{2cm}}$.	2	1,3,4	b,c,d	1,2

PART-B

Sl.No	QUESTION	M	BT	PO	CO																																													
1	a) Design a 4-bit shift register with a parallel load using D flip-flops. There are two control inputs: shift and load. When shift = 1, the contents of the register are shifted by one position right. New data is transferred into the register when load = 1 and shift=0. If both control inputs are equal to 0 the contents of the register do not change. b) Show the realization of a ring counter with 6 states.	6 4	2,3 4	a,b	1,2																																													
2	a) Design a serial subtractor using a sequential logic procedure. (JK FFs). Draw the state table and the state diagram for the same.	10	1,2 3	b,c, d,e	1,2 4, 5																																													
3	a) Design an arithmetic circuit with one selection variable S and two data inputs A and B. When S=0, circuit should performs the addition operation $F=A+B$. When S=1, the circuit should performs the increment operation $F=A+1$. b) Draw the logic diagram of a 4-bit combinational logic shifter.	6 4	1,2 3	a,b, d	1,2 4																																													
4	Design an arithmetic and logic circuit with three selection variables s_2, s_1, and s_0, that generates the following operations. Draw the logic diagram of one typical stage. Arithmetic operations: <table border="1"> <thead> <tr> <th>s_2</th><th>s_1</th><th>s_0</th><th>$C_{in}=0$</th><th>$C_{in}=1$</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td><td>$F=A$</td><td>$F=A+1$</td></tr> <tr> <td>0</td><td>0</td><td>1</td><td>$F=A+B$</td><td>$F=A+B+1$</td></tr> <tr> <td>0</td><td>1</td><td>0</td><td>$F=A-B-1$</td><td>$F=A-B$</td></tr> <tr> <td>0</td><td>1</td><td>1</td><td>$F=A-1$</td><td>$F=A$</td></tr> </tbody> </table> Logical operations: <table border="1"> <thead> <tr> <th>s_2</th><th>s_1</th><th>s_0</th><th>$C_{in}=0$</th></tr> </thead> <tbody> <tr> <td>1</td><td>0</td><td>0</td><td>$F=A \vee B$</td></tr> <tr> <td>1</td><td>0</td><td>1</td><td>$F=A \oplus B$</td></tr> <tr> <td>1</td><td>1</td><td>0</td><td>$F=A \wedge B$</td></tr> <tr> <td>1</td><td>1</td><td>1</td><td>$F=A$</td></tr> </tbody> </table>	s_2	s_1	s_0	$C_{in}=0$	$C_{in}=1$	0	0	0	$F=A$	$F=A+1$	0	0	1	$F=A+B$	$F=A+B+1$	0	1	0	$F=A-B-1$	$F=A-B$	0	1	1	$F=A-1$	$F=A$	s_2	s_1	s_0	$C_{in}=0$	1	0	0	$F=A \vee B$	1	0	1	$F=A \oplus B$	1	1	0	$F=A \wedge B$	1	1	1	$F=A$	10	4,5	a,b, c	1,2 3
s_2	s_1	s_0	$C_{in}=0$	$C_{in}=1$																																														
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1	1	1	$F=A$																																															
5	a) Describe with a suitable diagram the bus organization of a processor register with ALU. b) Write a note on status register.	6 4	1,2 4, 5	a,b,	1,2 4																																													

Department of Electronics and Communication Engineering

Subject title: Analysis and Design of Digital Circuits

Subject code:(12EC34)

November 11

 $\text{Test} = 1$

Time 12hours

Maximum Marks:65

Instructions to candidates:

1. Answer all the questions from Part A. Part A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B.

PART-A

Sl.No	QUESTION	M	BT	CO
1	Realize the function $F = (A + B')(C'D + E)$ using NAND gates only.	2	L1	CO1
2	Write the dual of the function $F = AB + A'C + BC + AB + A'C$.	1	L1	CO1
3	The circuit shown in the Fig. 1 is a _____ to _____ code converter.	1	L2	CO2
4	Identify essential prime implicants in the function given below. $F = \sum m(0, 2, 4, 7, 8, 10, 11, 12, 14, 15)$	1	L3	CO1
5	In Fig.2, logic expression(X) is _____.	2	L3	CO2
6	The minimum number of 2:1 multiplexers required for realizing an 8:1 multiplexer is _____.	1	L1	CO2
7	A 74147 priority encoder has ten active-LOW inputs and four active-LOW outputs. What would be the state of the four outputs, if inputs 4 and 5 are LOW and all other inputs are HIGH?	1	L3	CO3
8	In a binary multiplier, for J multiplier bits and K multiplicand bits, hardware requirement is _____ number of AND gates and _____ numbers of K bit adders.	1	L1	CO1
9	The decoders constructed using NAND gates are _____ term generators.	1	L1	CO1
10	If the input to a T-flipflop is a 100 Hz signal, the frequency of the final output of the three T-flipflops in cascade is _____.	2	L3	CO3
11	Realize 3X8 decoder using 2X4 decoders.	2	L2	CO2

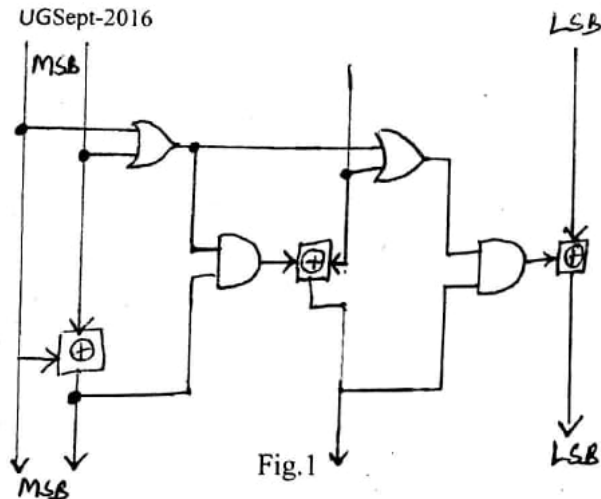


Fig.1

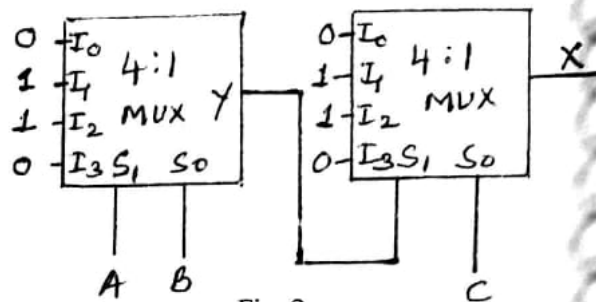


Fig. 2

PART-B

Sl.No	QUESTION	M	BT	CO
1 a)	Simplify the following Boolean expression using K-map method. $F = \sum m(0,2,3,5,7,15,18,21,22,26,27,28,29,31) + d(4,11,16,17)$	5	L3	CO1
b)	Design a code converter that converts a decimal digit from (8421) code to BCD using suitable ICs.	5	L5	CO2
2	Design a 4-bit carry look ahead adder and list out its advantages over a ripple carry adder.	10	L3	CO2
3	Implement the following two output functions (f_1 and f_2) using suitable decoder and external gates. $f_1 = (y' + x)z$ and $f_2 = (y' z' + x'y + yz')$	10	L5	CO2
4	Obtain the Boolean function $f(a,b,c,d) = \sum m(1,5,7,8,9,13,14)$ using suitable multiplexer with a,b and c as control inputs.	10	L4	CO1
5 a)	An automobile alarm circuit is used to detect certain undesirable conditions. There are three switches used to indicate the status of the door adjacent to driver's seat, the ignition and the head lights respectively. Design the logic circuit with these three switches as the inputs, so that the alarm will be activated whenever either of the following conditions exist: (a) The head lights are on while the ignition is off. (b) The door is open while the ignition is on.	5	L3	CO4
b)	Differentiate between sequential and combinational logic circuits. Derive the characteristic equation of SR-FF and JK-FF.	5	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	6	6	1	2	6	3	6	0	0	0
	Test	Max Marks	20	25	0	5	0	5	20	10	15	0

R.V.COLLEGE OF ENGINEERING, Bengaluru
 (Autonomous institution affiliated to VTU, Belagavi)
Department of Electronics and Communication Engineering

Semester: III

Test-II

Course Name: Analysis and Design of Digital Circuits
Time : 2 hours

Course Code: 12EC34

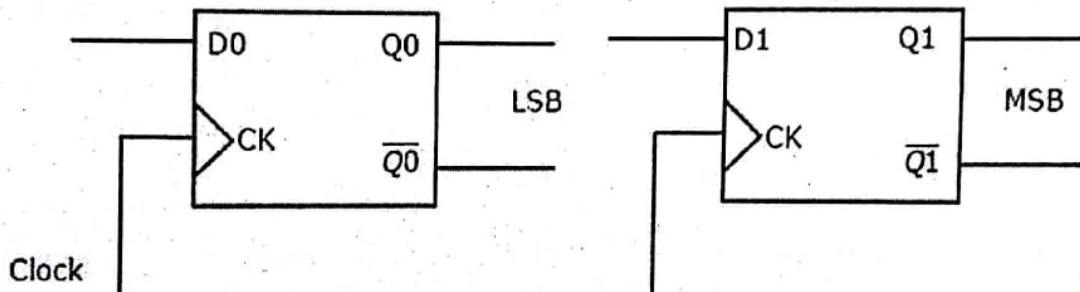
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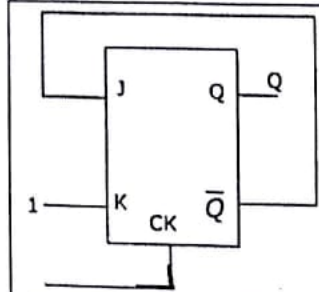
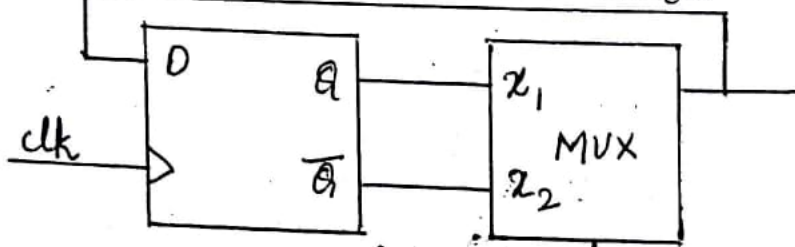
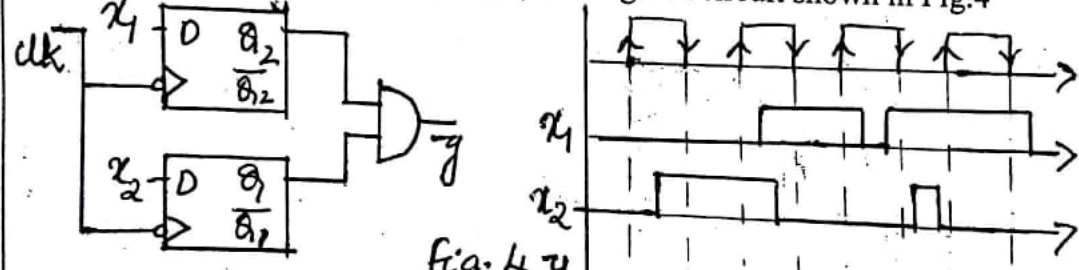
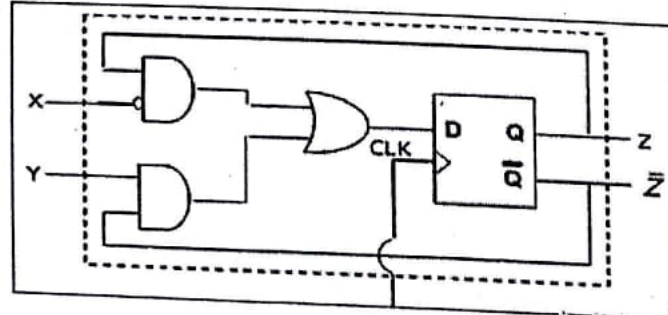
Maximum Marks: 65

Instructions to candidates:

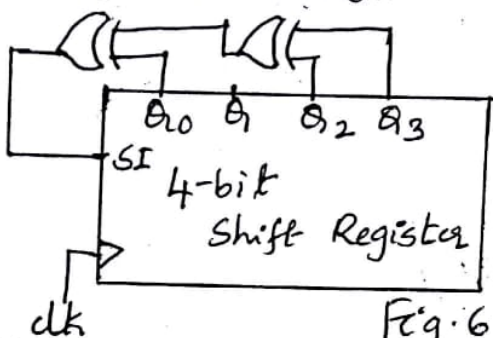
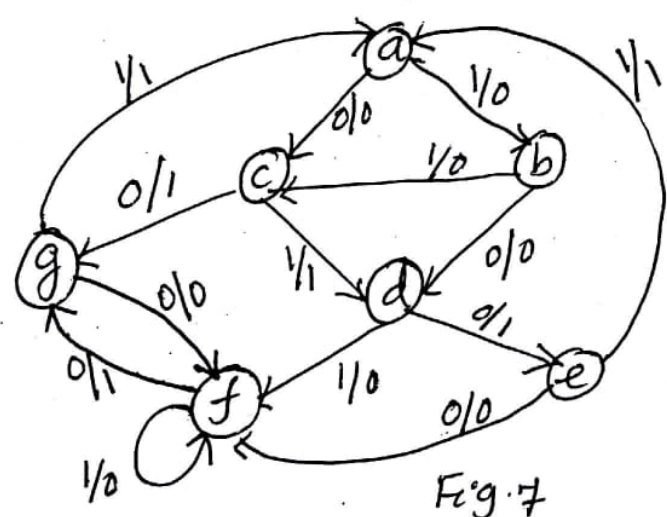
1. Answer all the questions from PART-A. PART-A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART-A.

PART-A

Sl.No	QUESTION	M	BT	CO
1	Two D flip-flops are connected as a synchronous counter as shown in Fig.1. The counter goes through the Q_1Q_0 sequence 00→01→11→10→00→..... The inputs D_0 and D_1 respectively are _____ & _____.  Fig.1	2	4	4
2	A K-bit Johnson counter produces _____ number of timing signals.	1	1	3
3	Define 'set up' and 'hold time' with respect to edge triggered flip flops.	1	1	3
4	List any two methods to eliminate race around condition in a JK FF.	1	2	1
5	_____ number of flip flops are complemented in a 10 bit binary ripple counter after the following count 1001100111.	1	4	2
6	Determine f_{max} for a 4-bit synchronous counter if, t_{pd} for each flip flop is 50ns and t_{pd} for each and gate is 20ns. Compare this with f_{max} for a mod-16 ripple counter.	2	3	4

7	In a JK flip-flop shown in Fig.2, $J=Q'$ and $K=1$. Assume the flip flop was initially cleared and then clocked for 6 pulses. The sequence at the Q output will be _____.  Fig.2	2	4	3
8	Draw the state diagram for the logic circuit shown in Fig.3.  Fig.3	2	5	1
9	Draw the timing diagram of output (Y) in the given circuit shown in Fig.4  Fig.4	1	3	2
10	Using IC-74193, design a down counter which counts from 1101 to 0011(13-3).	1	4	2
11	A sequential circuit using a D flip flop and logic gates is shown in Fig.5. X and Y are the inputs and Z is output. The circuit is _____.  Fig.5	1	2	1

PART-B

Sl.No	QUESTION	M	BT	CO
1 a)	A 4-bit shift register which shifts 1-bit to the right at every clock pulse, is initialized the pattern (1000) for ($Q_0Q_1Q_2Q_3$). The D input is derived from Q_0, Q_2 and Q_3 through XOR gates as shown in Fig.6.  Fig.6 i) Write the 4-bit pattern ($Q_0 Q_1 Q_2 Q_3$) after each clock pulse till pattern (1000) reappears on ($Q_0 Q_1 Q_2 Q_3$). ii) To what value should the shift register be initialized so that the pattern (1001) occurs after the first clock pulse.	5	4	3
1 b)	Design a positive edge triggered D-FF and explain its operation.	5	3	3
2	Design a clocked sequential circuit for the state diagram shown in Fig.7, using T-FFs. Also draw the reduced state diagram.  Fig.7	10	5	4
3 a)	Design an Asynchronous BCD counter using JK-FFs. Assume Preset and Clear inputs are disabled permanently.	6	3	1

3 b)	For the given set of operations, design a 4-bit universal shift register. <table><tr><td>Mode</td><td>operation</td></tr><tr><td>00</td><td>Hold</td></tr><tr><td>01</td><td>Shift Left</td></tr><tr><td>10</td><td>Shift Right</td></tr><tr><td>11</td><td>Complement the content</td></tr></table>	Mode	operation	00	Hold	01	Shift Left	10	Shift Right	11	Complement the content	4	2	1
Mode	operation													
00	Hold													
01	Shift Left													
10	Shift Right													
11	Complement the content													
4 a)	Design a sequential circuit with JK FF to satisfy the following state equations: $A(t+1) = A'B'CD + A'B'C + AC'D'$ $B(t+1) = A'C + CD' + A'BC'$ $C(t+1) = B$ $D(t+1) = D'$.	5	4											
4 b)	Construct a Johnson counter with 10 timing signals. Compare Johnson and Ring counters.	5	3	2										
5 a)	Describe the working of a JK master slave flip-flop with timing diagrams.	5	2	2										
5 b)	Draw the block diagram of a 4-bit serial adder using a 2-bit counter which outputs signal K after 11 state is reached. When start signal N is received, the register should be loaded. Assume that N=1 until addition is complete. When the addition is complete, the control network should go to a stop state and remain there until N is changed back to 0.	5	4	4										

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L
	Quiz	Max Marks	4	3	4	4	2	2	3	6	2	
	Test	Max Marks	15	10	10	15	-	9	16	15	10	

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R.V.COLLEGE OF ENGINEERING, Bengaluru-560059

(Autonomous institution affiliated to VTU, Belagavi)

Department of Electronics and Communication Engineering**Semester:III****Test-III****Course Name: Analysis and Design of Digital Circuits****Time :2hours****Course Code: 12EC34****Date: 2/11/2016****Maximum Marks:65****Instructions to candidates:**

1. Answer all the questions from PART-A. PART-A questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART-B.

PART-A

SLNo	QUESTION	M	BT	CO
1	The fastest Digital IC logic family is _____.	1	1	2
2	A logic gate in a particular logic family has following specifications: $V_{th} = 2$ volts, $V_{OH} = 4$ volts, $V_{IH} = 3$ volts, $V_{OL} = 1$ volt, and $V_{IL} = 1.5$ volts. Its noise margin is _____.	2	3	3
3	Define Propagation delay and Figure of merit.	1	1	2
4	The logic family which has the lowest power dissipation and lowest speed is _____.	1	1	1
5	If the output high current of a gate is 500mA and its input high current is 50mA, the fan out of the gate is _____.	1	3	2
6	What is the function performed by a parallel adder with inputs A and B with all bits of B made as logic 1 with $C_{in}=1$?	1	3	4
7	The status register condition after the operation of unsigned subtraction ($A-B$), if $A>B$ is _____.	1	2	3
8	Realize AND gate using Transmission gate.	2	4	3
9	A processor unit employs a scratchpad memory and it consists of 128 registers of eight bits each. i) How many lines are needed for address? ii) How many lines are there for the input data?	2	2	1
10	In the status register of 4 bit ALU, the result of a signed operation sets the V bit, if the result exceeds _____.	1	3	4
11	Design an inverter(NOT) gate using nMOS logic.	2	4	2

PART-B

SLNo	QUESTION	M	BT	CO																																																						
1.	With a neat diagram, explain the operation of NOR OR gate using Emitter Coupled Logic (ECL). Also mention the advantages and disadvantages of ECL.	10	2	2																																																						
2	Design an arithmetic and logic circuit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.	10	5	4																																																						
	<table border="1"> <thead> <tr> <th colspan="5">Arithmetic Operations</th><th colspan="4">Logical Operations</th></tr> <tr> <th>S_2</th><th>S_1</th><th>S_0</th><th>$C_{in}=0$</th><th>$C_{in}=1$</th><th>S_2</th><th>S_1</th><th>S_0</th><th>$C_{in}=X$</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td><td>$F=A$</td><td>$F=A+1$</td><td>1</td><td>0</td><td>0</td><td>OR</td></tr> <tr> <td>0</td><td>0</td><td>1</td><td>$F=A+B$</td><td>$F=A+B+1$</td><td>1</td><td>0</td><td>1</td><td>NOR</td></tr> <tr> <td>0</td><td>1</td><td>0</td><td>$F=A-B-1$</td><td>$F=A-B-1$</td><td>1</td><td>1</td><td>0</td><td>AND</td></tr> <tr> <td>0</td><td>1</td><td>1</td><td>$F=A-1$</td><td>$F=A$</td><td>1</td><td>1</td><td>1</td><td>Complement</td></tr> </tbody> </table>	Arithmetic Operations					Logical Operations				S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$	0	0	0	$F=A$	$F=A+1$	1	0	0	OR	0	0	1	$F=A+B$	$F=A+B+1$	1	0	1	NOR	0	1	0	$F=A-B-1$	$F=A-B-1$	1	1	0	AND	0	1	1	$F=A-1$	$F=A$	1	1	1	Complement			
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0	1	1	$F=A-1$	$F=A$	1	1	1	Complement																																																		
3 a)	Draw and explain the working of NAND and NOR gates using CMOS logic. Also mention the advantages of CMOS logic.	6	3	3																																																						
3 b)	Write a short note on status register of a processor.	4	1	1																																																						
4 a)	A 74LS00 NAND gate output is driving three 74LS00 NAND gates and two 74AS00 NAND gates. Determine if there is a loading problem. The specifications for the same are:	4	3	2																																																						
	<table border="1"> <thead> <tr> <th></th><th>74LS00</th><th>74AS00</th></tr> </thead> <tbody> <tr> <td>I_{OH}</td><td>-0.4mA</td><td>-2mA</td></tr> <tr> <td>I_{IH}</td><td>20μA</td><td>20μA</td></tr> <tr> <td>I_{OL}</td><td>8mA</td><td>20mA</td></tr> <tr> <td>I_{IL}</td><td>-0.4mA</td><td>-0.5mA</td></tr> </tbody> </table>		74LS00	74AS00	I_{OH}	-0.4mA	-2mA	I_{IH}	20 μ A	20 μ A	I_{OL}	8mA	20mA	I_{IL}	-0.4mA	-0.5mA																																										
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4 b)	What is a scratch pad memory? Explain the organization of a processor unit using scratch pad memory with the help of a neat diagram.	6	2	1																																																						
5	For the following set of micro operations, write control word register content. i) $R1 \leftarrow Shr R2$ ii) $R2 \leftarrow Shl R6$ iii) $R1 \leftarrow Crc R2$ iv) $R1 \leftarrow R5 - R7$ v) $R3 \leftarrow R5 xor R5$.	10	4	3																																																						

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	3	5	5	2	3	3	5	4	-	-
	Test	Max Marks	8	16	16	10	4	14	12	10	10	-

2013

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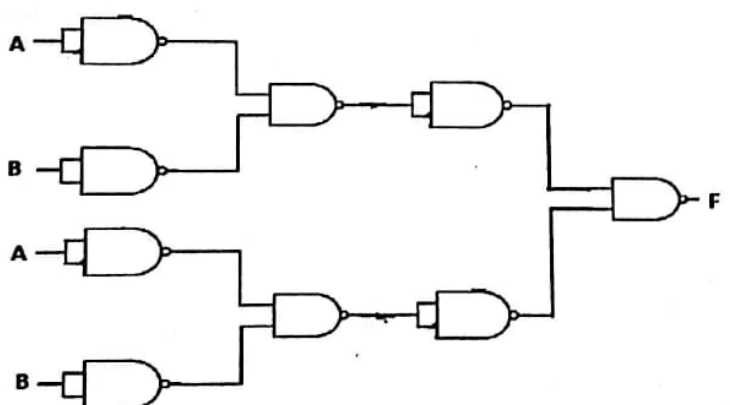
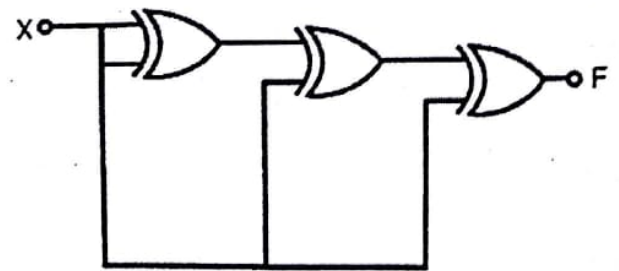
Analysis and Design of Digital circuits 12EC34
III Semester
1st Test

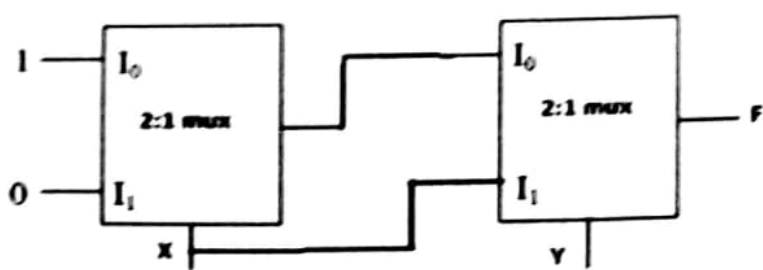
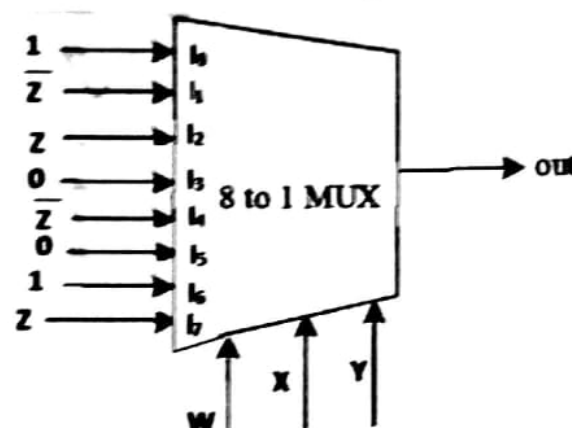
Time: 2 hours

Instructions to candidates:

Maximum Marks: 65

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B
- 3.

PART-A				
Sl.No	QUESTION	M	BT	CO
1	$(ABCD.3E)_{16} = \underline{\hspace{2cm}} (8).$	1	L1	CO1
2	Write the decimal number 493 in the following codes: i. 8421 code = <u> </u> . ii. 5211 code = <u> </u> .	2	L2	CO2
3	Given that $16_{10} = 100_b$, the value of b = <u> </u> .	1	L2	CO1
4	If the I_{OH} and I_{IH} of a gate are 500mA 50mA respectively, the fan-out is <u> </u> .	1	L2	CO1
5	For the circuit shown in fig.1, the output F is equal to <u> </u> .  fig1	1	L2	CO2
6	For the circuit shown in fig2, the output F is given by <u> </u> .  fig2	1	L2	CO2

7	For a digital system with three inputs A, B and C and one output Z, the output should be high when at least two inputs are high. The minimal expression for the output is	2	L1
8	In a full adder, SUM expression has _____ essential prime implicants and CARRY expression has _____ essential prime implicants.	2	L2
9	Find the functionality "F" of the circuit in fig3.	2	L2
 fig3			
10	What is the boolean function realized by the 8:1MUX shown in fig 4.	2	L2
 fig4			

PART B

	QUESTION	M	BT															
1	a) With a diagram, explain the operation of a CMOS NOR gate. b) Define noise margin? Calculate the noise margins of the TTL and CMOS gates describe below. <table><tr><td></td><td>CMOS</td><td>TTL</td></tr><tr><td>$V_{OH\ min}$</td><td>4.4V</td><td>2.4V</td></tr><tr><td>$V_{IH\ min}$</td><td>3.5V</td><td>2V</td></tr><tr><td>$V_{OL\ max}$</td><td>0.33V</td><td>0.4V</td></tr><tr><td>$V_{IL\ max}$</td><td>1.5V</td><td>0.8V</td></tr></table>		CMOS	TTL	$V_{OH\ min}$	4.4V	2.4V	$V_{IH\ min}$	3.5V	2V	$V_{OL\ max}$	0.33V	0.4V	$V_{IL\ max}$	1.5V	0.8V	5 5	L1 L2
	CMOS	TTL																
$V_{OH\ min}$	4.4V	2.4V																
$V_{IH\ min}$	3.5V	2V																
$V_{OL\ max}$	0.33V	0.4V																
$V_{IL\ max}$	1.5V	0.8V																
2	a) Simplify the Boolean expression, F, and show its implementation using NAND gates. $F(A,B,C,D,E) = \sum m(1,3,7,8,12,13,14,15,18,19,24,25,29)$ b) Describe the operation of an ECL OR gate with a diagram.	7 3	L3 L1															

a) Design a half subtractor circuit with inputs x and y and outputs D_{diff} and B_{out} . The circuit subtracts y from x and places the difference in D_{diff} and borrows in B_{out} .	4	L5	CO2
		L5	CO3
b) Design a 4-bit carry- look- ahead adder.	6		
a) Implement the following Boolean function using an 8:1 multiplexer. $F(A,B,C,D)=\sum m(1, 3, 4, 11, 12, 13,14,15)$	4	L5	CO2
b) Construct 4-to-16 line decoder with five 2-to-4 line decoder having enable inputs.	6	L5	CO3
a) Design a logic circuit for 2x2 binary multiplier using AND gates and binary adders.	5	L5	CO4
b) Write the truth table of an octal-to-binary priority encoder with MSB input having the highest priority. Draw its block diagram having an output V to indicate that at least one of the inputs is present. What will be the value of the four outputs, if inputs D_0 , D_2 and D_6 are at 1 at the same time?	5	L6	CO4

R.V.COLLEGE OF ENGINEERING, Bangalore
 (Autonomous institution affiliated to VTU)
Department of Electronics and Communication Engineering
ANALYSIS AND DESIGN OF DIGITAL CIRCUITS
 III Semester
 II Test

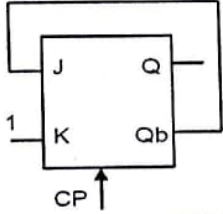
Time : 2 hours

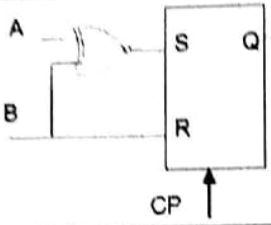
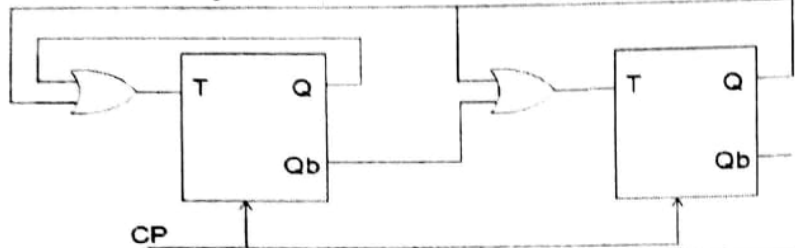
Maximum Marks:65

Instructions to candidates:

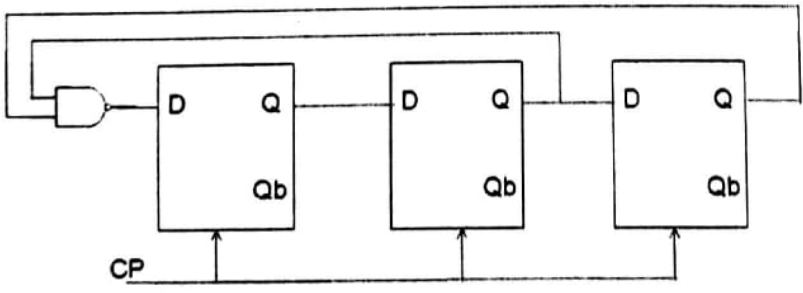
1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from Part B.

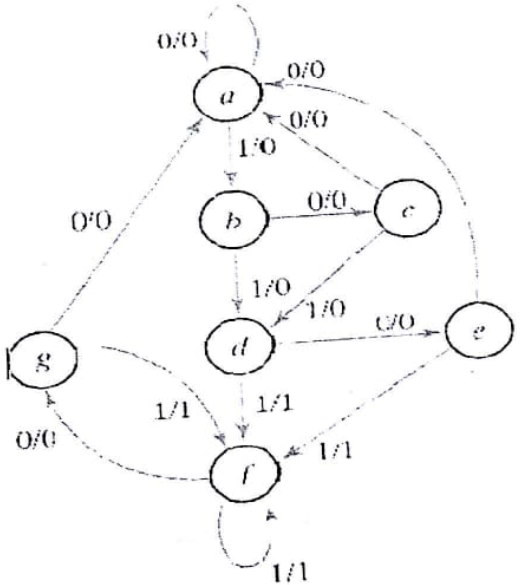
PART-A

Sl.No	QUESTION	Marks	BT	CO
1	What are the asynchronous inputs of a flip flop?	1	L1	CO1
2	What is the difference between asynchronous and synchronous counters?	1	L2	CO1
3	The Q output of SR flip flop is 1. The output does not change when clock pulse is applied. The inputs S and R may be respectively _____ and _____.	1	L2	CO2
4	The time for which input logic level must be held constant after the application of clock is _____.	1	L2	CO2
5	A JK flip flop is connected as shown below. Assuming the flip flop was initially cleared and then clocked for 8 pulses. The sequence at Q output will be _____. 	1	L3	CO3
6	A sequential circuit consists of 3 FFs A,B,C and 2 inputs x and y. The FF C defined by the state equation as given below is implemented using JK flip flop. What are the the input functions of FF? $C(t+1)=Cx+AB$	2	L4	CO3
7	The characteristic equation for SR flip flop is _____.	1	L1	CO1
8	A ripple counter consists of 8 FFs, each with a propagation delay of 10nS. What is the maximum clock frequency of the counter for reliable operation?	1	L3	CO4
9	The number of FFs required to build a binary counter circuit to count from 0 to 1023 is _____.	1	L3	CO4

10	Give an application each for flip-flops and counters?	1	L1	CO1
11	What is Race around condition? How can it be avoided?	1	L2	CO1
12	AB flip flop is constructed from SR flip flop as shown. The expression for next state is _____.	1	L4	CO3
				
13	The sequential circuit shown below consists of 2 T-FFs with initial state 00. What is the operation of the circuit?	2	L4	CO3
				

PART-B

SLNo	QUESTION	Marks	BT	CO
1 a)	Describe working of a JK master slave flip-flop with timing diagrams.	05	L3	CO1
b)	Obtain state table, state diagram and state equation for the sequential circuit in fig1b.	05	L4	CO3
 Fig1b				
2 a)	Show the realization of a mod-90 counter using 7490 decade counter.	04	L4	CO4
b)	Design mod-5 asynchronous counter using JK flip flop and write timing waveforms.	06	L6	CO2

3 a)	What are the different triggering methods used in flip-flops?	05	L2	CO1
b)	Convert a clocked SR flip flop to a clocked JK flip flop.	05	L3	CO1
4 a)	Show the realization of a synchronous 4 bit binary up-down counter using JK flip flops. Show the timing diagram for down counting operation.	07	L5	CO4
b)	Differentiate between Mealy and Moore circuits with an example each.	03	L1	CO2
5	For the state diagram shown in fig5, obtain the reduced state diagram. For the reduced state diagram, obtain state table. Also design the Digital circuit.  <pre> graph TD a((a)) -- "0/0" --> a a -- "0/0" --> c a -- "0/0" --> e a -- "1/0" --> b b -- "0/0" --> c b -- "1/0" --> d c -- "1/0" --> d c -- "0/0" --> e d -- "0/0" --> e d -- "1/1" --> f e -- "1/1" --> f f -- "1/1" --> f f -- "1/1" --> d f -- "0/0" --> g g -- "0/0" --> a g -- "1/1" --> f </pre> Fig 5	10	L5	CO4

NOV 2015

R.V.COLLEGE OF ENGINEERING, Bangalore
(Autonomous institution affiliated to VTU)
Department of Electronics and Communication Engineering

Analysis and Design of Digital circuits 12EC34
III Semester
3rd Test

Time: 2 hours

Instructions to candidates:

Maximum Marks: 65

1. Answer all the questions from Part A. Part A Questions should be answered in the first 2 pages of the answer book only.
2. Answer all questions from PART B

PART-A																			
Sl.No	QUESTION	M	BT	CO															
1	What is a register? What are its applications?	2	L1	CO1															
2	What is a Universal shift register?	2	L2	CO1															
3	The contents of a 4 bit shifter is initially 1101. The register is shifted six times to the right, with the serial input being 101101. What is the content of the register after each shift?	2	L3	CO2															
4	Why a Ring counter is called a 'counter'?	1	L2	CO2															
5	What is the difference between a Johnson counter and a Ring counter?	1	L2	CO2															
6	A serial adder uses two 4 bit registers A & B. Register A holds a value 0101 and register B holds a value 0111. The carry flip-flop Q is initially cleared. List the binary values in register A and flip-flop Q after each shift.	2	L2	CO3															
7	Why there is a difference between the designs of arithmetic unit and logical unit?	2	L1	CO3															
8	The contents of a flag register VZSC after an operation of ALU is 1011. What do this mean?	2	L2	CO4															
9	Which combinational circuit is used to generate Z flag?	1	L2	CO4															
PART B																			
	QUESTION	M	BT	CO															
1	a) Design a 4 bit Universal shift register and draw the circuit with the given mode of operation table. <table><tr><td>S₁</td><td>S₀</td><td>Operation</td></tr><tr><td>0</td><td>0</td><td>Parallel</td></tr><tr><td>0</td><td>1</td><td>Shift Right</td></tr><tr><td>1</td><td>0</td><td>Shift Left</td></tr><tr><td>1</td><td>1</td><td>Inhibit Clock</td></tr></table>	S ₁	S ₀	Operation	0	0	Parallel	0	1	Shift Right	1	0	Shift Left	1	1	Inhibit Clock	10	L5	CO4
S ₁	S ₀	Operation																	
0	0	Parallel																	
0	1	Shift Right																	
1	0	Shift Left																	
1	1	Inhibit Clock																	

2	a) Draw the circuit which will implement a 4 bit serial in, parallel out shift register and explain its working.	5	L3	CO2
	b) Construct a Johnson counter using D flip-flops. Specify the AND gate outputs required to output the decimal numbers 0 to 8.	5	L4	CO3
3	a) Design a 4 bit Adder/ Subtractor circuit with one selection variable S and two inputs A&B. When S=0, this circuit performs A+B. When S=1, it performs the operation A-B taking two's complement of B.	10	L5	CO4
4	a) Design a logic circuit for CPU using MUX and OR, XOR, AND & NOT gates.	5	L4	CO2
	b) Mention one method of integrating Arithmetic circuit and Logical circuit inside CPU, with block diagram and explanation.	5	L4	CO3
5	a) To an 8 bit ALU, a four bit status register is attached. How to set/reset the condition bits V, Z, S & C?	5	L5	CO4
	b) Give the diagram of a 4 bit combinational logic shifter with MUXs and explain its working.	5	L3	CO4
